

LIKELY AND UNLIKELY RAMIFICATION OF ALGEBRAIC POINTS ON CURVES

BIANCA VIRAY AND ISABEL VOGT

ABSTRACT. This paper initiates the study of arithmetic properties of closed points on curves of arbitrary degree. We show that a pointless curve of rank 0 can have only finitely many cyclic degree ℓ points for *all* sufficiently large primes ℓ . More generally, we show that the residue fields of points on curves of genus at least 2 and rank 0 are biased away from total ramification. The proof technique illuminates a natural intermediate conjecture between the Manin–Mumford conjecture and the Zilber–Pink conjecture for self-products of abelian varieties. This intermediate conjecture implies analogous results for other splitting profiles.

1. INTRODUCTION

A central focus in arithmetic geometry is understanding how the geometry of a variety influences its arithmetic. For example, the Mordell conjecture (proved by Faltings) asserts that curves of genus at least 2 can have only finitely many points defined over any fixed number field k . In other words, for curves of genus at least 2, the map

$$\{x \in C : x \text{ closed}\} \rightarrow \{L/k : [L:k] < \infty\}, \quad x \mapsto \mathbf{k}(x) \quad (1.1)$$

sending a point x to its residue field $\mathbf{k}(x)$ is finite-to-1. (Equivalently, this map sends a Galois orbit of geometric points to the isomorphism class of a minimal field of definition of a point in the orbit.) One can also ask about the image of the map, and the philosophy that geometry controls arithmetic suggests that for a curve of genus at least 2 the image of (1.1) should be constrained.

In this project we explore this question and discover that, for many curves, large ramification degrees are unlikely to appear in residue fields of closed points. For example, we prove the following.

Theorem 1.1. *Let C/k be a nice curve whose Jacobian has rank 0. If $C(k) = \emptyset$ and there are no (geometric) points of total ramification in each k -rational map $f:C \rightarrow Z$ with Z a curve of genus 0 and $\deg(f) = \text{index}(C/k)$, then there exists an integer m such that for every d coprime to m , total ramification is **unlikely** for degree $d \cdot \text{index}(C/k)$ points on C . Precisely, the set of places*

$$\{v \in \Omega_k : \exists x \in C, \deg(x) = d \cdot \text{index}(C/k), \mathbf{k}(x)/k \text{ totally ramified at } v\}$$

is finite.

If we further restrict to points on C of *prime* degree, Theorem 1.1 has the following consequence for cyclic points.

Corollary 1.2. *Let C/k be a nice curve whose Jacobian has rank 0. If $C(k) = \emptyset$, then for all sufficiently large primes ℓ , there are finitely many cyclic degree ℓ points on C .*

BV was partially supported by NSF DMS-2401327. IV was partially supported by NSF DMS-2338345.

Proof. The conclusion holds vacuously if $\text{index}(C/k) > 1$, since then there are no points of large prime degree. We may therefore assume that $\text{index}(C/k) = 1$. The assumptions mean that we may apply Theorem 1.1 to conclude that for all primes ℓ coprime to m (in particular, all sufficiently large primes), total ramification for degree ℓ points can only occur at a finite set of places $\mathcal{T}_\ell$. In a cyclic extension of prime degree, any ramified prime must be totally ramified. Thus, if $\mathbf{k}(x)$ is cyclic of degree $\ell > 2$ it must be contained in the ray class field of modulus $\prod_{v \in \mathcal{T}_\ell} v^{n_v}$, where $n_v = 1$ if $v \nmid \ell$ and $n_v = 2$ if $v \mid \ell$. This ray class field is a number field, so by Faltings's (Curve) Theorem there are finitely many points of C defined over this ray class field. $\square$

Remarks 1.3.

- (i) If $\text{genus}(C) \leq 1$, then every degree d divisor is linearly equivalent to dp , for a geometric point $p \in C(\bar{k})$. Hence, the assumptions of Theorem 1.1 force $\text{genus}(C) \geq 2$.
- (ii) If $C(k) \neq \emptyset$ or if there exists a map $C \rightarrow \mathbb{P}^1$ of degree $\text{index}(C/k)$ that is totally ramified at some geometric point, then the set of places in Theorem 1.1 has positive density for all sufficiently large d (see Corollary 4.9 and Theorem 1.11).
- (iii) For many curves satisfying the assumptions of Theorem 1.1, e.g., curve 7676.a.7676.1, we can show that the theorem holds with $m = 2$, so odd total ramification is *always* unlikely. (In particular, for these curves, “sufficiently large primes ℓ ” in Corollary 1.2 means ℓ odd.) For small odd degrees ($e = 3, 5$), we can compute the finite set of places of good reduction where degree e points can be totally ramified; see Table 1.1.

We obtain results analogous to Theorem 1.1 for other ramification indices. In order to precisely state our results in this direction, we introduce some notation.

Definition 1.4.

- (i) A **splitting profile** $\mathbf{P}$ is a finite multiset of ordered pairs $\mathbf{P} = \{(e_1, f_1), \dots, (e_r, f_r)\}$. Define $r(\mathbf{P}) := |\mathbf{P}|$ and $\deg(\mathbf{P}) := \sum_{i=1}^r e_i f_i$. From a splitting profile $\mathbf{P}$, we can make two length $r(\mathbf{P})$ ordered tuples, well-defined up to simultaneous permutation:

$$\mathbf{e}(\mathbf{P}) := (e_1, \dots, e_r), \quad \mathbf{f}(\mathbf{P}) := (f_1, \dots, f_r).$$

- (ii) A splitting profile $\mathbf{P}$ is **unramified** if $\mathbf{e}(\mathbf{P}) = (1, \dots, 1)$, and **ramified** otherwise.
- (iii) Given a finite extension L/k and a nonarchimedean place v of k , the ramification and residual degrees of the places lying over v give a splitting profile of degree $[L : k]$:

$$\mathbf{P}_v(L/k) := \{(e(L_w/k_v), [\mathbb{F}_w : \mathbb{F}_v]) : w|v\}.$$

- (iv) A splitting profile $\mathbf{P}$ is **likely** in V , a set of effective divisors on C , if

$$\#\{v : \exists x \in V \text{ integral, with } \mathbf{P}_v(\mathbf{k}(x)/k) = \mathbf{P}\} = \infty,$$

and **unlikely** otherwise. We say that $\mathbf{P}$ is **likely/unlikely** on C when V is taken to be a set that contains all closed points of degree equal to $\deg(\mathbf{P})$.

Remarks 1.5.

- (i) If $\mathbf{P}$ is an unramified splitting profile and there exists a point $x \in C$ and a place v such that $\mathbf{P}_v(\mathbf{k}(x)/k) = \mathbf{P}$, then $\mathbf{P}$ is likely in any set containing x by the Chebotarev density theorem.
- (ii) If the set $\{\mathbf{k}(x) : x \in V \text{ integral}\}$ is finite and $\mathbf{P}$ is a ramified splitting profile, then $\mathbf{P}$ is unlikely in V .

- (iii) Since k is dense in $\prod_{v \in S} k_v$ for any finite set of places S , there exist monic irreducible polynomials over k that have prescribed v -adic factorization at any finite set of places v . Therefore, given a splitting profile $\mathbf{P}$ and a finite set of places S , one can construct a number field L/k with $\mathbf{P}_v(L/k) = \mathbf{P}$ for all $v \in S$. For example, an extension totally ramified at all places in S can be obtained by adjoining a root of a polynomial $f(x)$ that is Eisenstein at v for all $v \in S$. Since every number field occurs as the residue field of a closed point on $\mathbb{P}^1$, every splitting profile is likely on $\mathbb{P}^1$.
- (iv) If a splitting profile $\mathbf{P}^0$ is likely on a curve C , then every sufficiently large multiple of $\mathbf{P}^0$, i.e., $d\mathbf{P}^0 := \{(de, f) : (e, f) \in \mathbf{P}^0\}$, is likely on C (Proposition 4.7.)

Theorem 1.6. *Let C/k be a nice curve whose Jacobian has rank 0, and let $\mathbf{P}^0 = \{(e_i, f_i)\}_i$ be a splitting profile such that $\text{index}(C) | e_i f_i$ for all i and such that $\frac{\deg(\mathbf{P}^0)}{\text{index}(C)}$ is relatively prime to $\#J_C(k)$. If C has no points of degree at most $\deg(\mathbf{P}^0)$, then there exists an integer $m = m(C, \mathbf{P}^0)$ such that for every integer d coprime to m , the splitting profile $\mathbf{P} = d\mathbf{P}^0 = \{(de_i, f_i)\}_i$ is **unlikely** on C .*

Remark 1.7. Theorem 1.1 can be rephrased as saying if $\mathbf{P}^0 := \{(\text{index}(C), 1)\}$ is unlikely, then arbitrarily large multiples $d\mathbf{P}^0 := \{(de, f) : (e, f) \in \mathbf{P}^0\}$ are also unlikely. A similar rephrasing of Theorem 1.6 is possible, but one instead must assume that $\mathbf{P}^0$ and all of its refinements are unlikely. See Section 5 for more details.

A key input to the proofs of Theorems 1.1 and 1.6 is the Manin-Mumford conjecture (proved by Raynaud). We obtain similar statements for splitting profiles where the ramification indices are allowed to vary over a group of rank at least 2, conditional on a generalization of the Manin-Mumford conjecture (also a specialization of the Zilber-Pink conjecture).

Theorem 1.8. *Let C/k be a nice curve whose Jacobian has rank 0 and is geometrically simple. Let $\mathbf{P}^0 = \{(e_i^0, f_i)\}$ be a splitting profile. Assume:*

- *the Zilber–Pink conjecture (or the weaker Conjecture 2.3),*
- $\sum_i f_i < g(C)$,
- $\text{index}(C) | e_i^0 f_i$ for all i ,
- $\min_i(e_i^0 f_i) / \text{index}(C)$ is relatively prime to $\#J_C(k)$, and
- *there are no points of degree at most $\min_i(e_i^0 f_i)$ on C .*

*Then, for all $\mathbf{e} \in \prod_i e_i^0 \mathbb{Z}_{>0}$ outside of a proper finite union of subgroups, the splitting profile $\mathbf{P} = \{(e_i, f_i)\}_i$ is **unlikely** on C .*

Remark 1.9. Since $\prod_i e_i \mathbb{Z}_{>0}$ is finitely generated, the complement of a proper finite union of subgroups must include a coset of a finite index subgroup [VV26, Lemma 4.3.8]. If $J_C(k) = \{O\}$ and $\text{index}(C) = 1$, then for each $\mathbf{f} = (f_1, \dots, f_r)$ for which $\sum_i f_i < g$ and C has no points of degree $\min_i(f_i)$, Theorem 1.8 therefore implies that positive density set of splitting profiles $\mathbf{P}$ with $\mathbf{f}(\mathbf{P}) = \mathbf{f}$ are unlikely.

As is typically expected in the unlikely intersections framework, there are some geometric conditions that can cause a splitting profile to be likely. The assumptions that $\sum_i f_i < g(C)$, and that there are no points of degree at most $\min_i(e_i^0 f_i)$ exclude these scenarios. If the Jacobian of C has nontrivial isogeny factors then the assumptions required are more delicate. For the most general version of Theorem 1.8, see Theorem 3.7.

The methods of this paper apply more generally to determine whether particular splitting profiles are unlikely in the linear systems $\{|dD|\}_{d \in \mathbb{Z}_{>0}}$ for D a fixed divisor of positive degree (See Section 3 for more details on this). For a curve C/k that has index 1 and whose Jacobian has rank 0, the union of linear systems

$$\bigcup_{D \in \text{Pic}_C^1(k)} \bigcup_{d \in \mathbb{Z}_{>0}} |dD|$$

contains all closed points on C whose degree is coprime to $\#J_C(k)$, and thus we can deduce statements like Theorems 1.1, 1.6, and 1.8.

The method of proof also gives an effective (but not necessarily efficient) method to compute the finite set of places where an unlikely splitting profile can be realized. We compute these finite sets for 20 genus 2 curves in the case of total ramification in degrees 3 and 5; they are listed in Table 1.1. The proof that Table 1.1 is correct is in Section 3.2.3 and the associated code is in the [Github repository](#) for this paper. (We compute these finite sets using a short Weierstrass model for C , so this model has bad reduction at 2.)

LMFDB label	$\{p > 2 : \exists x \in C, P_p(\mathbf{k}(x)/k) = P, \text{ and } C \text{ has good reduction at } p\}$	
	$P = \{(3, 1)\}$	$P = \{(5, 1)\}$
7676.a.7676.1		316004230293431
9977.a.9977.1	2027	2837
11259.a.11259.1		232695223753
13072.a.13072.1		10301, 23625124907
15504.a.62016.1		11, 6451166627922673
19170.c.76680.1	35153	
21989.a.21989.1	3	1685914369
22032.a.22032.1		7054572663424903
22185.a.22185.1		4903622393
23033.a.23033.1		1277, 19181, 492731
23735.a.23735.1		225889
25707.a.77121.1		199, 28882395713187018090626537
29636.a.474176.1		6719
31174.a.342914.1		1044737, 553763689903, 10546179748889
31668.a.63336.1		11
36114.a.36114.1		5, 1091
41151.a.41151.1		41, 131, 293, 12647, 175891
43335.b.43335.1		17, 6583714939
45175.a.45175.1	3	88339, 5807603, 95190432083453372821
47449.a.47449.1		431, 2939, 880057

TABLE 1.1. Genus 2 curves that satisfy Theorem 1.1 with $m = 2$, so odd total ramification is unlikely. For $e = 3, 5$, we list the finite set of odd primes of good reduction that are totally ramified in the residue field of some degree e point on C . See Section 3.2.3 for more details.

The results mentioned thus far focus on criteria for when a giving splitting profile is unlikely. We also explore the complimentary question of when splitting profiles are likely.

For example, we show analogously to the case when $C = \mathbb{P}^1$ mentioned in Remark 1.5(iii), all splitting profiles are likely on an elliptic curve.

Proposition 1.10. *Let E be an elliptic curve over a number field k . For any $x \in E(k)$ and any splitting profile P , the set*

$$\{v \in \Omega_k : \exists y \in E, y \sim \deg(P)x, P_v(\mathbf{k}(y)/k) = P\}$$

contains all sufficiently large primes. In particular, every splitting profile with $\deg(P) = d$ is likely in the complete linear system $|dx|$.

The key ingredient in the proof is a lifting result, which may be of independent interest (Theorem 4.1): given an effective divisor over $\mathbb{F}_v$ in a sufficiently positive linear system, we construct a closed point $x \in C_k$ whose splitting profile is determined by the components of the effective divisor over $\mathbb{F}_v$. This lifting result also allows us to show in some situations that a likely splitting profile is realized by a positive density set of places. For example, in the context of Theorem 1.1 we prove the following.

Theorem 1.11. *Let C/k be a nice curve whose Jacobian has rank 0. For any $d \geq 2g(C) + 1$, the set of places*

$$\{v \in \Omega_k : \exists x \in C, \deg(x) = d, \mathbf{k}(x)/k \text{ totally ramified at } v\}$$

is Frobenian. In particular, if the set is infinite, it has positive density.

See Corollary 5.6 for the most general version of this statement.

1.1. Related work.

1.1.1. ℓ -Diophantine Stability. In [MR18], Mazur and Rubin introduce the notion of ℓ -Diophantine stability: given a prime integer ℓ , a variety X over a global field k is ℓ -Diophantine-stable if for every n , there are infinitely many cyclic extensions L/k of degree ℓ^n such that X has no points whose residue fields are contained in L (in other words, $X(L) = X(k)$). Among other results, they prove that every curve of genus at least 1 whose endomorphism ring is equal to its geometric endomorphism ring is ℓ -Diophantine-stable for a positive density set of ℓ [MR18, Theorem 1.2]. This result is similar, but not directly comparable, to Corollary 1.2, even for a fixed curve that satisfies both hypotheses. Indeed, Corollary 1.2 gives a stronger restriction on cyclic points of *prime* degree than [MR18, Theorem 1.2]. However, our results do not constrain cyclic points with degree a nontrivial power of a prime, since cyclic extensions of degree ℓ^r with $r > 1$ can have ramified primes that are not totally ramified. On the other hand, [MR18, Theorem 1.2] rules out infinitely many degree ℓ^r cyclic residue fields for a positive proportion of primes ℓ .

Remark 1.12. If one is interested in studying closed points on a curve C whose residue field is Galois with Galois group G , then the relevant moduli space is $C^{|G|}/G$. If G is cyclic of order at least 3 and C has genus at least 2, then $C^{|G|}/G$ is general type [Raw24, Theorem 1], so the Bombieri-Lang conjecture predicts that the rational points are not Zariski dense.

1.1.2. *Unlikely intersections.* As mentioned above, a key input in the proof of Theorem 1.6 is the Manin–Mumford conjecture (proved by Raynaud), which describes the structure of $Z \cap A_{\text{tors}}$, where $Z \subset A$ is a closed subvariety of an abelian variety. For the more general, albeit conditional, Theorem 1.8, the relevant intersection is

$$Z \cap \bigcup_{\mathbf{e}} A^r[\mathbf{e}],$$

where the union ranges over some subset of positive integer vectors $\mathbf{e} \in \mathbb{Z}_{>0}^r$, $Z \subset A^r$ is a closed subvariety of a self-product of an abelian variety, and $A^r[\mathbf{e}]$ denotes the kernel of the morphism $A^r \xrightarrow{[\mathbf{e}]} A$, $(p_1, \dots, p_r) \mapsto \sum_i e_i p_i$. Specializing the Zilber–Pink conjecture to intersections of this form gives Conjecture 2.3, which we contend is a natural intermediate generalization of the Manin–Mumford conjecture. (The case when the union ranges over a rank 1 subgroup of $\mathbb{Z}^r$ can be reduced to the Manin–Mumford conjecture.) To the best of our knowledge, Conjecture 2.3 is open, even in the simplest case where we obtain interesting applications: when $\dim A = 3$, $r = 2 = \dim Z$ and A is geometrically simple.

1.1.3. *Splitting profiles in basepoint free pencils.* The problem of determining which splitting profiles appear as $P_v(\mathbf{k}(x)/k)$ for x ranging over a basepoint free pencil, i.e., fibers of a morphism $C \rightarrow \mathbb{P}^1$, has been explored extensively for Galois morphisms. Indeed, works of Beckmann, Dèbes, Ghazi, König, Legrand, and Neftin [Bec91, DG12, DL13, KLN19, Leg17] give a complete answer to this question (for all but finitely many places), and explore its applications to the Grunwald and parameterization problems. In the non-Galois case, recent work of Balçık, Chan, Liu, and the first author [BCLV] study the splitting profiles $P_v(\mathbf{k}(C_t)/k)$ and the isomorphism classes of $\mathbf{k}(C_t) \otimes_k k_v$ for all but finitely many places. The results in these papers give necessary and sufficient conditions for a splitting profile to be realized for all places where the morphism $C \rightarrow \mathbb{P}^1$ has good reduction. This may exclude finitely many primes in addition to the primes where the curve has bad reduction.

1.2. **Outline.** We begin in Section 2 setting up the unlikely intersections framework needed to prove our main theorems. As described above, the intersections relevant to our problem are of the form

$$Z \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r} A^r[\mathbf{e}],$$

where $Z \subset A^r$ is a closed subvariety of a self-product of an abelian variety, and $A^r[\mathbf{e}]$ denotes the kernel of the morphism $A^r \xrightarrow{[\mathbf{e}]} A$, $(p_1, \dots, p_r) \mapsto \sum_i e_i p_i$. While the Zilber–Pink conjecture puts forth a necessary criterion for this intersection to be Zariski dense in Z , we need to extract from this a condition on $\mathbf{e}$ that guarantees that the intersection $Z \cap A^r[\mathbf{e}]$ is *empty*. The key is to give a complete characterization of how non-empty intersections arise in the case when A is simple (Proposition 2.11) that we can then inductively leverage to show that away from finitely many subgroups with rank strictly less than r , nonempty intersections arise from torsion translates that have larger than expected intersection (Theorem 2.12). These results may be of independent interest.

In Section 3, we lay out the general strategy to determine when splitting profiles are unlikely in a complete linear system. The most general versions of our main theorems about unlikely splitting profiles are Theorems 3.5 and 3.7. Theorems 1.1, 1.6 and 1.8 are obtained as corollaries of Theorems 3.5 and 3.7 as explained in Sections 3.2.1 and 3.2.2. In Section 3.2.3,

we explain how we computed the information in Table 1.1 and raise questions about finer properties of the sets of places realizing total ramification.

In Section 4, we explore converses to the results in Section 3 for very ample linear systems. The main theorem of this section is Theorem 4.1, which morally can be viewed as an extension of Hensel's Lemma and Hilbert irreducibility to the case of non-smooth subschemes. From this theorem, we give several simple conditions that each guarantee likely ramification (Section 4.3) which together give a proof of Proposition 1.10 (Section 4.3.3).

We close in Section 5 with a study of the set of places realizing a given splitting profile, including the proof of Theorem 1.11.

Remark 1.13. For readability, we have endeavored to make the sections as independent as possible. For example, Sections 2 and 4 can each be read independently of all other sections. A reader willing to accept Proposition 2.1 and Theorem 2.5 may read Section 3 independently of all other sections. Additionally, while Section 3.1 provides helpful context for Section 5, the arguments within Section 5 depend only on Theorem 4.1.

1.3. Notation. We use k to denote a number field. For any finite set of places S , we write $\mathcal{O}_{k,S} \subset k$ for the ring of S -integers. Given a nonarchimedean place v , we write $\mathbb{F}_v$ for the residue field of v .

A nice curve C is a smooth projective geometrically integral scheme of dimension 1 over a field. Given a nice curve C , we write $g(C)$ for its genus, Pic_C for its Picard scheme, and J_C for the identity component (its Jacobian). Given a divisor D on C we write $[D]$ to denote its divisor class (modulo linear equivalence) and, given another divisor E on C , we write $E \sim D$ if $[E] = [D]$. We write $|D|$ for the complete linear system of D . More generally, for any subvariety $W \subset \text{Pic}_C^d$, we write $|W|$ to denote the set of effective divisors whose linear equivalence class is contained in W .

For a positive integer f , we write Sym_C^f for the f^{th} symmetric product and $W_C^f \subset \text{Pic}_C^f$ for the locus of effective degree f divisors. Given a vector of positive integers $\mathbf{f} \in \mathbb{Z}_{>0}^r$, we write $W^{\mathbf{f}}$ for the product $W^{f_1} \times \cdots \times W^{f_r}$. For any divisor D of positive degree, translation by a suitable multiple of D gives an isomorphism between a component of the Picard scheme containing divisors with degree divisible by $\deg(D)$ and the identity component J_C .

More generally, if $\mathbf{f}$ is a tuple and D is a divisor of positive degree with $\deg(D) \mid f_i$ for all i , then we write ι_D for the morphism

$$\text{Pic}_C^{f_1} \times \cdots \times \text{Pic}_C^{f_r} \xrightarrow{\sim} J_C^r, \quad (z_1, \dots, z_r) \mapsto \left(z_1 - \frac{f_1}{\deg(D)} D, \dots, z_r - \frac{f_r}{\deg(D)} D \right).$$

Given an abelian group A and a tuple of positive integers $\mathbf{w} \in \mathbb{Z}_{>0}^r$, componentwise multiplication (alternatively multiplication by a diagonal matrix with integer entries) gives a morphism that we denote

$$[\text{Diag}(\mathbf{w})]: A^r \rightarrow A^r, \quad (z_1, \dots, z_r) \mapsto (w_1 z_1, \dots, w_r z_r).$$

If A is torsionfree (e.g., $A = \mathbb{Z}$), then $[\text{Diag}(\mathbf{w})]$ is injective. Under this assumption, given an element $\mathbf{w}' \in [\text{Diag}(\mathbf{w})]A^r$, we write $\mathbf{w}'/\mathbf{w}$ for the unique preimage $[\text{Diag}(\mathbf{w})]^{-1}(\mathbf{w}')$. We write $\sigma: A^r \rightarrow A$ for the summation map $\sigma(z_1, \dots, z_r) = z_1 + \cdots + z_r$, and we write $[\mathbf{w}]$ for the composition $\sigma \circ [\text{Diag}(\mathbf{w})]$, which can be thought of as a dot product map. We denote the kernel of this morphism by $A^r[\mathbf{w}]$.

Given a smooth integral model $\mathcal{C} \rightarrow \text{Spec } \mathcal{O}_{k,S}$ with generic fiber $C_k = C$, the relative Picard variety $\text{Pic}_{\mathcal{C}/\mathcal{O}_{k,S}}$ is then a group scheme over $\text{Spec } \mathcal{O}_{k,S}$ and every component $\text{Pic}_{\mathcal{C}/\mathcal{O}_{k,S}}^d$ is

smooth and proper over $\mathrm{Spec} \mathcal{O}_{k,S}$. Given a closed subscheme $H \subset \mathrm{Pic}_{C/k}$, we write $H_{\mathbb{F}_v}$ for the fiber of the closure of H in $\mathrm{Pic}_{C/\mathcal{O}_{k,S}}$ over v . (This is nonempty by the valuative criterion for properness.) In particular, there exists a unique limit of any k -divisor class D modulo v for every $v \in \mathrm{Spec} \mathcal{O}_{k,S}$, which we denote $D_{\mathbb{F}_v}$. A closed point $x \in C$ determines a map $\mathrm{Spec} \mathcal{O}_{k(x)} \rightarrow C$.

ACKNOWLEDGEMENTS

We thank Laura Capuano, Aaron Levin, Joe Silverman, and Melanie Matchett Wood for conversations that helped contextualize these results.

2. A GENERALIZATION OF THE MANIN–MUMFORD CONJECTURE

The questions and results considered here are geometric, so in this section we work over $\mathbb{C}$. The Manin–Mumford Conjecture (now a theorem of Raynaud) states that the torsion points on a subvariety $Z \subset A$ of an abelian variety A decomposes as a finite union of torsion translates contained in Z [Ray83a, Théorème I, ; Ray83b, Théorème, page 327]. As a consequence, if Z does not contain 0, then $Z \cap A[m] = \emptyset$ for a positive density set of positive integers.

Proposition 2.1. *Let A be an abelian variety and let $Z \subset A$ be a closed subvariety such that $0 \notin Z$. Then there exist positive integers $m_1, \dots, m_s \in \mathbb{Z}_{\geq 2}$ such that*

$$Z \cap A[m] \neq \emptyset \iff m \in \bigcup_i m_i \mathbb{Z}.$$

(See Section 2.2 for the proof.)

We are interested in generalizations where we consider a subvariety of a power of an abelian variety, $Z \subset A^r$ and its intersection with the subgroups $A^r[\mathbf{e}]$ defined above. (This case arises in our study of unlikely splitting profiles in linear systems of curves; see Section 3.1 for details about this connection.)

Informally, the Zilber–Pink conjecture predicts that Z contains a Zariski dense set of points that are in $\bigcup_{\mathbf{e}} A^r[\mathbf{e}]$ only if codimension estimates predict that it is likely. More precisely, we have the following.

Definition 2.2. If $Z \subset A^r$ is an integral subvariety, let $\langle Z \rangle \subset A^r$ be the smallest algebraic subgroup of A^r containing Z and define

$$\mathcal{E}_Z := \{\mathbf{e} \in \mathbb{Z}^r : \dim \langle Z \rangle \cap A^r[\mathbf{e}] \geq \dim \langle Z \rangle - \dim Z\}. \quad (2.1)$$

Conjecture 2.3 (Follows from the Zilber–Pink Conjecture; see Section 2.5 for details). *Let A be an abelian variety and let $Z \subset A^r$ be an integral subvariety. Then*

$$Z \cap \left(\bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}] \right)$$

is not Zariski dense in Z .

Remark 2.4. Observe that when $r = 1$, the subgroup $A^r[\mathbf{e}]$ is simply the e_1 -torsion $A[e_1]$ and so $\dim \langle Z \rangle \cap A[e_1] \leq 0$. Additionally, $\dim Z = \dim \langle Z \rangle$ if and only if Z is a torsion translate. Hence, when $r = 1$, $\mathcal{E}_Z \neq \emptyset$ if and only if Z is a torsion translate and therefore the above conjecture reduces to the Manin–Mumford conjecture.

Assuming this conjecture, we can prove a generalization of Proposition 2.1 to $r > 1$.

Theorem 2.5. *Assume Conjecture 2.3. Let A be an abelian variety and let $Z \subset A^r$ be a subvariety. Suppose that there exists $\tilde{\mathbf{e}} \in \mathbb{Z}^r$ such that for every abelian subvariety $B \subset A$ and every torsion point $\tau \in A^r$, either $\dim Z \cap (\tau + B^r) < \dim B$ or $[\tilde{\mathbf{e}}]\tau \notin B$. Then there exist finitely many proper subgroups $M_1, \dots, M_s \subsetneq \mathbb{Z}^r$ such that*

$$Z \cap \left(\bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \bigcup_i M_i} A^r[\mathbf{e}] \right) = \emptyset.$$

The subgroups may be chosen so that additionally $\bigcup_{i=1}^s M_i \subsetneq \mathbb{Z}^r$ and $\tilde{\mathbf{e}} \notin M_i$ for each finite index subgroup M_i

Remarks 2.6.

- (i) As mentioned in the introduction, once $\bigcup_i M_i \subsetneq \mathbb{Z}^r$, the complement $\mathbb{Z}^r \setminus \bigcup_i M_i$ must include a coset of a finite index subgroup [VV26, Lemma 4.3.8].
- (ii) An $\tilde{\mathbf{e}}$ satisfying the assumptions of Theorem 2.5 will never exist if $0 \in Z$, (the condition fails for $\tau = \mathbf{0}, B = \{0\}$). The conclusions also fail in this case since $0 \in A^r[\mathbf{e}]$ for all $\mathbf{e} \in \mathbb{Z}^r$. More generally, if $\dim Z \cap B^r \geq \dim B$ for any abelian subvariety B (including if $\dim Z \cap A^r = \dim Z \geq \dim A$), then such an $\tilde{\mathbf{e}}$ can not exist. See Examples 2.7 and (ii) below, which illustrate why this condition is necessary.
- (iii) Without the assumption on the existence of $\tilde{\mathbf{e}}$, we prove that for $\mathbf{e}$ outside a finite union of subgroups of rank at most $r - 1$, the intersection $Z \cap \bigcup A^r[\mathbf{e}]$ is constrained to lie in finitely many torsion translates (Theorem 2.12). Without assuming the existence of $\tilde{\mathbf{e}}$, this result combines with Lemma 2.8 below to show that as long as $\dim Z \cap B^r < \dim B$ for all abelian subvarieties $B \subseteq A$, there are proper subgroups $M_1, \dots, M_s$ such that $Z \cap \bigcup_{\mathbf{e} \notin \bigcup_i M_i} A^r[\mathbf{e}] = \emptyset$. However, even though each $M_i \neq \mathbb{Z}^r$, the union $\bigcup M_i$ may cover $\mathbb{Z}^r$. This can indeed happen. Let E be an elliptic curve and let $\tau \in E[2] \setminus \{0\}$ be a nontrivial 2-torsion point. Let $Z \subset E^2$ be an integral curve containing the torsion points $T_1 := (\tau, 0)$, $T_2 := (0, \tau)$, and $T_3 := (\tau, \tau)$, and not containing $(0, 0)$. Then $T_1 \in E^2[\mathbf{e}]$ if and only if e_1 is even, $T_2 \in E^2[\mathbf{e}]$ if and only if e_2 is even, and $T_3 \in E^2[\mathbf{e}]$ if and only if e_1 and e_2 have the same parity. Hence $Z \cap E^2[\mathbf{e}]$ is nonempty for every $\mathbf{e} \in \mathbb{Z}^r$.

Examples 2.7.

- (i) Let C be a nice genus 2 curve and let $D \in \text{Pic}_C^1 \setminus W_C^1$ be an ineffective divisor of degree 1. Let $Z_C := \iota_D(W_C^{(1,1)}) \subset J_C^2$. Since $D \notin W_C^1$, we have $0 \notin Z_C$. On the other hand, for any $\mathbf{e} \in \mathbb{Z}^2$ with $e_1 e_2 \neq 0$, the restriction of $[\mathbf{e}]$ to Z_C gives a surjective map $Z_C \rightarrow J_C$. This implies that $Z_C \cap J_C^2[\mathbf{e}] \neq \emptyset$. (Indeed, since the torsion in J_C is dense, we additionally have that $Z_C \cap \bigcup_m J_C^2[m\mathbf{e}]$ is Zariski dense in Z_C .) In particular, the conclusion of Theorem 2.5 fails. This does not contradict Theorem 2.5 because $\dim Z_C = 2 = \dim J_C$, and so no $\tilde{\mathbf{e}}$ can exist (Remark 2.6(ii)).
- (ii) Retain the setup from the previous example. In addition, let X be a smooth curve with a morphism ϕ of degree f to the genus 2 curve C . Assume that X has genus at least $2f + 1$. Let $A := J_X$ and let $B := \phi^* J_C$. Assume that D is such that $\phi^* D \in \text{Pic}_X^f$ is also ineffective. Let $Z := \iota_{\phi^* D}(W_X^{(f,f)}) \subset A^2$. Our assumption that $\phi^* D \notin W_X^f$ guarantees that $0 \notin Z$. Since $\dim Z = 2f$, our assumption that X has genus at least $2f + 1$ guarantees that $\dim Z < \dim A$.

We have a commutative diagram

$$\begin{array}{ccc} \iota_D \left(W_C^{(1,1)} \right) = Z_C & \hookrightarrow & J_C^2 \\ \downarrow \phi^* & & \downarrow \phi^* \\ \iota_{\phi^* D} \left(W_X^{(f,f)} \right) = Z & \hookrightarrow & J_X^2 = A^2. \end{array}$$

Since $B^2[\mathbf{e}] \subset A^2[\mathbf{e}]$, we have $\phi^*(Z_C \cap J_C^2[\mathbf{e}]) = \phi^*(Z_C) \cap B^2[\mathbf{e}] \subset Z \cap A^2[\mathbf{e}]$. In particular, for any $\mathbf{e}$ for which $e_1 e_2 \neq 0$, we have $Z \cap A^2[\mathbf{e}] \neq \emptyset$. This is consistent with Theorem 2.5 since $\dim(Z \cap B^2) \geq \dim Z_X = 2$ is not less than $\dim B = 2$, and so no such $\tilde{\mathbf{e}}$ can exist (Remark 2.6(ii)). $\square$

Outline of Section 2. If Z is a torsion translate, then we may prove Theorem 2.5 unconditionally; we do this in Section 2.1 (Lemma 2.8). In Section 2.2, we prove Proposition 2.1. In Section 2.3, we prove Theorem 2.5 in the case that A is simple. Then in Section 2.4 we prove Theorem 2.5 by induction on the number of simple factors in an isogeny decomposition. Finally, in Section 2.5, we show that the Zilber-Pink conjecture (Conjecture 2.14) implies Conjecture 2.3.

2.1. The case when Z is a nontrivial torsion translate.

Lemma 2.8. *Let $B \subset A$ be an abelian subvariety and let $\tau \in A_{\text{tors}}^r$. Then*

$$M_{\tau+B^r} := \{\mathbf{e} \in \mathbb{Z}^r : (\tau + B^r) \cap A^r[\mathbf{e}] \neq \emptyset\}$$

is a full rank subgroup of $\mathbb{Z}^r$, which is a proper subgroup if and only if $\tau \notin B^r$. Explicitly, when $r = 1$, we have $M_{\tau+B^r} = \text{ord}_{A/B}(\tau)\mathbb{Z}$.

Proof. By definition, $\mathbf{e} \in M_{\tau+B^r}$ if and only if $[\mathbf{e}](\tau + B) = [\mathbf{e}]\tau + B$ contains $0 \in A$. Hence $\mathbf{e} \in M_{\tau+B^r}$ if and only if $[\mathbf{e}]\tau \in B$. The set $M_{\tau+B}$ is thus a subgroup. Since $\tau \in A_{\text{tors}}^r$, for each $1 \leq i \leq r$, there exists a positive integer m_i such that $m_i \text{pr}_i(\tau) \in B$ (here pr_i denotes the i^{th} projection). In particular, for each $1 \leq i \leq r$, the vector $(0, \dots, 0, m_i, 0, \dots, 0)$ is contained in $M_{\tau+B^r}$. We conclude that the subgroup has full rank. If $\tau \notin B^r$, we know that some $\text{pr}_i(\tau) \notin B$, and so $(0, \dots, 0, 1, 0, \dots, 0) \notin M_{\tau+B^r}$. Thus, in this case, this subgroup is not all of $\mathbb{Z}^r$.

The explicit description when $r = 1$ follows from the definition $M_{\tau+B^r} = \{e \in \mathbb{Z} : e\tau \in B\}$. $\square$

2.2. Consequence of the Manin–Mumford Conjecture.

Theorem (Manin–Mumford Conjecture, proved by Raynaud [Ray83a, Théorème I; Ray83b, Théorème, page 327]). *Let $Z \subset A$ be a closed subvariety of an abelian variety. Then*

$$Z \cap A_{\text{tors}} = \bigcup_{i=1}^s \tau_i + (B_i)_{\text{tors}},$$

where for all i , $\tau_i \in A_{\text{tors}}$ and $B_i \subset A$ is an abelian subvariety such that $\tau_i + B_i \subset Z$.

Proof of Proposition 2.1. By treating each irreducible component separately (and taking the (finite) union of the corresponding lists of integers) the statement is equivalent to the case when Z is irreducible. We will prove this by induction on the dimension of Z . If $Z \cap A_{\text{tors}}$

is not Zariski dense in Z , then, replacing Z with each component of the Zariski closure of $Z \cap A_{\text{tors}}$, we are done by induction. If $Z \cap A_{\text{tors}}$ is Zariski dense in Z , then, by the Manin–Mumford conjecture, Z is a torsion translate $Z = \tau + B$ of an abelian subvariety $B < A$. Since $0 \notin Z$, we have $\tau \notin B$ and so $\text{ord}_{A/B}(\tau) > 1$. The result therefore follows from Lemma 2.8, letting $m_1 = \text{ord}_{A/B}(\tau)\mathbb{Z}$. $\square$

2.3. The simple case. When A is simple, we will show that the subgroups M_i in Theorem 2.5 can be recursively constructed as $\mathcal{E}_Z$ for Z an irreducible component of the Zariski closure of $Z \cap \bigcup_{\mathbf{e} \notin \bigcup_{j < i} M_j} A^r[\mathbf{e}]$ that is not a torsion point, union finitely many subgroups of the form M_τ for torsion points τ . First we prove that, when A is simple and $\dim Z < \dim A$, the sets $\mathcal{E}_Z$ are indeed subgroups with rank strictly less than r .

Lemma 2.9. *Let A be a simple abelian variety and let $Z \subset A^r$ be an integral subvariety with $\dim Z < \dim A$. If Z is not a torsion point, then $\mathcal{E}_Z$ (see Definition 2.2) is a subgroup of $\mathbb{Z}^r$ of rank strictly less than r .*

Proof. Since $A^r[\mathbf{e}]$ is the kernel of the morphism $[\mathbf{e}]: A^r \rightarrow A$, we have $\dim \langle Z \rangle \cap A^r[\mathbf{e}] = \dim \langle Z \rangle - \dim[\mathbf{e}]\langle Z \rangle$. Thus

$$\begin{aligned} \mathcal{E}_Z &:= \{\mathbf{e} : \dim \langle Z \rangle \cap A^r[\mathbf{e}] \geq \dim \langle Z \rangle - \dim Z\} \\ &= \{\mathbf{e} : \dim \langle Z \rangle - \dim[\mathbf{e}]\langle Z \rangle \geq \dim \langle Z \rangle - \dim Z\} \\ &= \{\mathbf{e} : \dim Z \geq \dim[\mathbf{e}]\langle Z \rangle\}. \end{aligned}$$

Since A is simple, $\dim[\mathbf{e}]\langle Z \rangle$ is equal to 0 or $\dim A$. By assumption, $\dim Z < \dim A$, so

$$\mathcal{E}_Z = \{\mathbf{e} : \dim Z \geq \dim[\mathbf{e}]\langle Z \rangle\} = \{\mathbf{e} : \dim[\mathbf{e}]\langle Z \rangle = 0\} = \{\mathbf{e} : \langle Z \rangle^\circ \subset A^r[\mathbf{e}]\},$$

where $\langle Z \rangle^\circ$ denotes the identity component of $\langle Z \rangle$. The set $\{\mathbf{e} : \langle Z \rangle^\circ \subset A^r[\mathbf{e}]\}$ is evidently a subgroup of $\mathbb{Z}^r$. Furthermore, by Lemma 2.10 below, $\text{rank } \mathcal{E}_Z = r$ if and only if $\dim \langle Z \rangle^\circ = 0$, which occurs if and only if Z is a torsion point, thus completing the proof. $\square$

Lemma 2.10. *For $\mathbb{Q}$ -linearly independent $\mathbf{e}_1, \dots, \mathbf{e}_r \in \mathbb{Z}^r$, the intersection $A[\mathbf{e}_1] \cap \dots \cap A[\mathbf{e}_r]$ is 0-dimensional.*

Proof. Let $E \in M_r(\mathbb{Z})$ be the matrix whose rows are $\mathbf{e}_1, \dots, \mathbf{e}_r$. Then $(z_1, \dots, z_r) \in A[\mathbf{e}_1] \cap \dots \cap A[\mathbf{e}_r]$ if and only if $E(z_1, \dots, z_r)^T = \mathbf{0}$. By assumption E is a matrix with full rank, so $\det(E) \neq 0$. If $E(z_1, \dots, z_r)^T = \mathbf{0}$, then

$$(\text{Adj}(E)E)(z_1, \dots, z_r)^T = (\det(E)z_1, \dots, \det(E)z_r)^T = \mathbf{0},$$

and so $A[\mathbf{e}_1] \cap A[\mathbf{e}_2] \cap \dots \cap A[\mathbf{e}_r] \subset A_{\text{tors}}$ is 0-dimensional. $\square$

Proposition 2.11. *Let A be a simple abelian variety and let $Z \subset A^r$ be an integral subvariety. Assume Conjecture 2.3 and that $\dim Z < \dim A$.*

- (1) *Then there exist subgroups $M_1, \dots, M_t$ of rank strictly less than r and a finite set of torsion points T such that*

$$Z \cap \bigcup_{\mathbf{e} \notin \bigcup_{i=1}^t M_i} A^r[\mathbf{e}] = T$$

(2) If there exists an $\tilde{\mathbf{e}} \in \mathbb{Z}^r \setminus \{0\}$ such that $Z \cap A_{\text{tors}}^r \cap A^r[\tilde{\mathbf{e}}] = \emptyset$, then there exist rank r proper subgroups $M_{t+1}, \dots, M_s$ with $\tilde{\mathbf{e}} \notin \cup_{i=t+1}^s M_i$ (which implies $\cup_{i=1}^s M_i \subsetneq \mathbb{Z}^r$) such that

$$Z \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^s M_i} A^r[\mathbf{e}] = \emptyset.$$

Proof. We will prove the first statement by induction on the dimension of Z . If Z is a torsion point, it trivially holds with $t = 0$ (no subgroups). If Z is a point of infinite order then, by Lemma 2.9, $\mathcal{E}_Z \subsetneq \mathbb{Z}^r$ is a subgroup of rank at most $r - 1$. By Conjecture 2.3,

$$Z \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}]$$

is not Zariski dense in Z , and hence is empty.

Now assume that Z has positive dimension. Again, by Lemma 2.9, $\mathcal{E}_Z \subsetneq \mathbb{Z}^r$ is a subgroup of rank at most $r - 1$ and by Conjecture 2.3,

$$Z \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}]$$

is not Zariski dense in Z . Thus, there exist $W_1, \dots, W_n \subset Z$, each integral and of strictly smaller dimension such that

$$Z \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}] = \bigcup_{j=1}^n \left(W_j \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}] \right).$$

So, by induction, there exists a finite collection of subgroups $M_1, \dots, M_{t-1}, M_t = \mathcal{E}_Z \subsetneq \mathbb{Z}^r$ of rank strictly less than r and a finite set of torsion points T such that

$$Z \cap \bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \cup_{i=1}^t M_i} A^r[\mathbf{e}] = T.$$

It remains to prove the second statement. Assume that there exists a nonzero $\tilde{\mathbf{e}}$ such that $Z \cap A_{\text{tors}}^r \cap A^r[\tilde{\mathbf{e}}] = \emptyset$. In particular, $0 \notin Z$. Thus, for each $\tau \in T$, the $B = 0$ case of Lemma 2.8 gives a proper rank r subgroup $M = M_\tau = M_{\tau+0^r}$ such that $\tau \notin Z \cap A^r[\mathbf{e}]$ if and only if $\mathbf{e} \notin M_\tau$. Let $M_{t+1}, \dots, M_s$ be the subgroups M_τ for $\tau \in T$. Then $Z \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^s M_i} A^r[\mathbf{e}] = \emptyset$ and $\tilde{\mathbf{e}} \notin \cup_{i=t+1}^s M_i$. A finite union of proper subgroups of $\mathbb{Z}^r$ that excludes a single element must exclude an entire coset of a finite index subgroup [VV26, Lemma 4.3.8]. Therefore the proper containment $\cup_{i=t+1}^s M_i \subsetneq \mathbb{Z}^r$ implies that $\cup_{i=1}^s M_i$ is also properly contained in $\mathbb{Z}^r$. $\square$

2.4. The general case and proof of Theorem 2.5.

Theorem 2.12. *Let A be an abelian variety and let $Z \subset A^r$ be a subvariety. Assume Conjecture 2.3 and that $\dim Z < \dim A$. There exist subgroups $M_1, \dots, M_t$ of rank strictly less than r and torsion cosets $\tau_1 + B_1^r, \dots, \tau_n + B_n^r$ with $\dim Z \cap (\tau_j + B_j^r) \geq \dim B_j$ such that*

$$Z \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^t M_i} A^r[\mathbf{e}] \subset \bigcup_{j=1}^n Z \cap (\tau_j + B_j^r).$$

If A is simple, then Proposition 2.11 shows there exists a finite union M of rank at most $r - 1$ subgroups such that $Z \cap \bigcup_{\mathbf{e} \notin M} A^r[\mathbf{e}]$ equals a finite set of torsion points. In the general case, it is not possible to upgrade the inclusion in Theorem 2.5 to an equality with a finite set of torsion translates (or even Zariski density in a finite set of torsion translates), as the following example illustrates.

Example 2.13. Let $Z_C \subset J_C^2$ be as in Example 2.7, so that for any fixed $\mathbf{e} \in \mathbb{Z}^2$ with $e_1 e_2 \neq 0$, we have $Z_C \cap \bigcup_{m \in \mathbb{Z}} J_C^2[m\mathbf{e}]$ is Zariski dense in Z_C . We may further assume that J_C is simple. Let $A := J_C \times A'$ for some simple positive-dimensional abelian variety A' , and consider $Z = Z_C \times \{\tau\}^2 \subset A^2 = J_C^2 \times A'^2$ for $\tau \in A'$ a torsion point. Then $2 = \dim Z < 2 + \dim A' = \dim A$. The torsion translate $\{0\}^2 \times \{\tau\}^2 + J_C^2 \times \{0\}^2$ contains Z , and so the theorem holds with $\tau_1 = \{0\}^2 \times \{\tau\}^2$ and $B_1 = J_C \times \{0\}$. The set of $\mathbf{e} \in \mathbb{Z}^2$ such that $[\mathbf{e}]\tau^2 = 0 \in A'$ is a proper subgroup of rank 2 in $\mathbb{Z}^2$ by Lemma 2.8. For any such $\mathbf{e} \in \mathbb{Z}^2$ with $e_1 e_2 \neq 0$, we have that the Zariski closure of $Z \cap \bigcup_{m \in \mathbb{Z}} A^2[m\mathbf{e}]$ is equal to $Z \subset \tau_1 + B_1^2$, which is not a torsion translate. $\square$

Proof of Theorem 2.12. The statement of the theorem when Z is integral formally implies the general statement, by considering each irreducible component of Z separately. We may therefore assume that Z is integral.

We induct on the number of simple factors of A . If A is simple this is Proposition 2.11(1). Now let $G_1, \dots, G_s \subset A$ be abelian subvarieties such that $A_i := A/G_i$ is simple and the product of quotient maps gives an isogeny $A \rightarrow A_1 \times \dots \times A_s$. Let $Z_i \subset A_i^r$ be $Z \bmod G_i^r$.

By assumption,

$$\dim Z_1 + \dots + \dim Z_s = \dim Z < \dim A = \dim A_1 + \dots + \dim A_s,$$

so there is some i such that $\dim Z_i < \dim A_i$; let $I \subset \{1, \dots, s\}$ be the set of all indices where $\dim Z_i < \dim A_i$. Write $A_I := \prod_{i \in I} A_i$, $Z_I := \prod_{i \in I} Z_i$, and $G_I := \ker(A \rightarrow A_I) = \cap_{i \in I} G_i$.

For each $i \in I$, we may apply Proposition 2.11 to $Z_i \subset A_i^r$ to obtain subgroups $M_{i,1}, \dots, M_{i,t_i}$ of rank at most $r-1$ and a finite set T_i of torsion points such that $Z_i \cap \bigcup_{\mathbf{e} \notin \bigcup_{j=1}^{t_i} M_{i,j}} A_i^r[\mathbf{e}] \subset T_i$. In particular,

$$T_I := Z_I \cap \left(\bigcup_{\mathbf{e} \notin \bigcup_{i \in I} \bigcup_{j=1}^{t_i} M_{i,j}} A_I^r[\mathbf{e}] \right) \subset \prod_{i \in I} T_i. \quad (2.2)$$

Let $\tilde{T}_I \subset A_{\text{tors}}^r$ be a set that maps bijectively to T_I under projection. Then (2.2) implies

$$Z \cap \left(\bigcup_{\mathbf{e} \notin \bigcup_{i \in I} \bigcup_{j \leq t_i} M_{i,j}} A^r[\mathbf{e}] \right) = \bigcup_{\tau \in \tilde{T}_I} Z \cap (\tau + G_I^r) \cap \left(\bigcup_{\mathbf{e} \notin \bigcup_{i \in I} \bigcup_{j \leq t_i} M_{i,j}} A^r[\mathbf{e}] \right).$$

Define the subset $\tilde{S}_I := \{\tau \in \tilde{T}_I : \dim Z \cap (\tau + G_I^r) = \dim(-\tau + Z) \cap G_I^r < \dim G_I\}$. For each $\tau \in \tilde{T}_I \setminus \tilde{S}_I$, we have that $\dim Z \cap (\tau + G_I^r) = \dim(-\tau + Z) \cap G_I^r \geq \dim G_I$. So the union over $\tau \in \tilde{T}_I \setminus \tilde{S}_I$ is trivially contained in a finite union of torsion cosets with the desired properties. Thus, to prove Theorem 2.12, it suffices to prove that for each $\tilde{\tau}$ in $\tilde{S}_I$, there exist subgroups $\tilde{M}_1, \dots, \tilde{M}_t$ of rank strictly less than r and torsion cosets $\tau_1 + B_1^r, \dots, \tau_n + B_n^r$ with $\dim Z \cap (\tau_j + B_j^r) \geq \dim B_j$ such that

$$Z \cap (\tilde{\tau} + G_I^r) \cap \left(\bigcup_{\mathbf{e} \notin \bigcup_i \tilde{M}_i} A^r[\mathbf{e}] \right) = \tilde{\tau} + \left[(-\tilde{\tau} + Z) \cap G_I^r \cap \left(\bigcup_{\mathbf{e} \notin \bigcup_i \tilde{M}_i} -\tilde{\tau} + A^r[\mathbf{e}] \right) \right] \subset \bigcup_j \tau_j + B_j^r. \quad (2.3)$$

Henceforth, we work with a fixed $\tilde{\tau} \in \tilde{S}_I$. Since G_I has fewer simple factors than A and $\dim -\tau + Z < \dim G_I$ by the definition of $\tilde{S}_I$, we may apply our inductive hypothesis to $(-\tilde{\tau} + Z) \cap G_I^r$. Hence, there exist rank at most $r-1$ subgroups $N_1, \dots, N_s \subsetneq \mathbb{Z}^r$ and torsion

translates $\sigma_1 + H_1^r, \dots, \sigma_m + H_m^r \subset G_I^r$ such that $\dim(-\tilde{\tau} + Z) \cap (\sigma_j + H_j^r) \geq \dim H_j$ and

$$(-\tilde{\tau} + Z) \cap G_I^r[\mathbf{e}] \subset \bigcup_{j=1}^m (-\tilde{\tau} + Z) \cap (\sigma_j + H_j^r) \quad \text{for all } \mathbf{e} \notin \cup_{\ell} N_{\ell}. \quad (2.4)$$

For any $w \in \mathbb{Z} \setminus \{0\}$ and any $\mathbf{e} \in \mathbb{Z}^r$, we have $A^r[\mathbf{e}] \subset A^r[w\mathbf{e}]$ by definition. Choose $w_0 \in \mathbb{Z} \setminus \{0\}$ such that $w_0\tilde{\tau} = \mathbf{0}$. Then,

$$G_I^r \cap (-\tilde{\tau} + A^r[\mathbf{e}]) \subset G_I^r \cap (-\tilde{\tau} + A^r[w_0\mathbf{e}]) = G_I^r[w_0\mathbf{e}]. \quad (2.5)$$

For each $i = 1, \dots, s$, let $\tilde{M}_i := \frac{1}{w_0} N_i = \{\mathbf{e} : w_0\mathbf{e} \in N_i\}$. By construction, $\text{rank } \tilde{M}_i = \text{rank } N_i$. Combining (2.5) and (2.4) we obtain

$$\begin{aligned} \tilde{\tau} + \left[(-\tilde{\tau} + Z) \cap G_I^r \cap \left(\bigcup_{\mathbf{e} \notin \cup_i \tilde{M}_i} -\tilde{\tau} + A^r[\mathbf{e}] \right) \right] &\subset \tilde{\tau} + \left[(-\tilde{\tau} + Z) \cap \bigcup_{\mathbf{e} \notin \cup_i \tilde{M}_i} G_I^r[w_0\mathbf{e}] \right] \\ &= \tilde{\tau} + \left[(-\tilde{\tau} + Z) \cap \bigcup_{\mathbf{e} \notin \cup_i N_i} G_I^r[\mathbf{e}] \right] \\ &\subset \tilde{\tau} + \left[\bigcup_{j=1}^m (-\tilde{\tau} + Z) \cap (\sigma_j + H_j^r) \right] \\ &= \bigcup_{j=1}^m Z \cap (\tilde{\tau} + \sigma_j + H_j^r). \end{aligned}$$

Since $\tilde{\tau} + \sigma_j$ is torsion for all j , this gives the desired containment as in (2.3). $\square$

Proof of Theorem 2.5. By Theorem 2.12, there exist subgroups $M_1, \dots, M_t$ of rank strictly less than r and torsion cosets $\tau_1 + B_1^r, \dots, \tau_n + B_n^r$ with $\dim Z \cap (\tau_j + B_j^r) \geq \dim B_j$ such that

$$Z \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^t M_i} A^r[\mathbf{e}] \subset \bigcup_{j=1}^n (\tau_j + B_j^r) \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^t M_i} A^r[\mathbf{e}].$$

By Lemma 2.8, for each $j \in \{1, \dots, n\}$, there is a full rank subgroup $M_{t+j} := M_{\tau_j + B_j} \subset \mathbb{Z}^r$ such that $(\tau_j + B_j^r) \cap A^r[\mathbf{e}] \neq \emptyset$ if and only if $\mathbf{e} \in M_{t+j}$. Excluding these subgroups we have

$$Z \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^{t+n} M_i} A^r[\mathbf{e}] \subset \bigcup_{j=1}^n (\tau_j + B_j^r) \cap \bigcup_{\mathbf{e} \notin \cup_{i=1}^{t+n} M_i} A^r[\mathbf{e}] = \emptyset.$$

Setting $s = t + n$, it therefore suffices to show that $\bigcup_{j=1}^{t+n} M_j \neq \mathbb{Z}^r$. By hypothesis, there exists $\tilde{\mathbf{e}} \in \mathbb{Z}^r$ such that $[\tilde{\mathbf{e}}]\tau_j \notin B_j$ for all j . Hence $\tilde{\mathbf{e}} \notin \bigcup_{j=1}^n M_{t+j}$. Since a finite union of subgroups omitting a single element omits an entire coset of a finite index subgroup [VV26, Lemma 4.3.8], and $M_1, \dots, M_t$ have rank at most $r - 1$, we conclude that $\bigcup_{j=1}^{t+n} M_j \neq \mathbb{Z}^r$. $\square$

2.5. Relationship to the Zilber–Pink conjecture.

Conjecture 2.14 (Zilber–Pink). *Let B be an abelian variety. Let $V \subset B$ be a subvariety that is not contained in a proper algebraic subgroup of B . Then the intersection*

$$V \cap \left(\bigcup_{\substack{H \subset B \text{ alg. subgroup} \\ \dim H + \dim V < \dim B}} H \right)$$

is not Zariski dense in Z .

Given an algebraic group H , we write H° for the connected component of the identity.

Lemma 2.15. *Let $H_1, H_2 \subset G$ be algebraic subgroups of an algebraic group. Then we have $\dim H_1 \cap H_2 = \dim H_1^\circ \cap H_2$.*

Proof. Any component of the intersection $H_1 \cap H_2$ is a torsor under the identity component $(H_1 \cap H_2)^\circ$, and so $\dim H_1 \cap H_2 = \dim(H_1 \cap H_2)^\circ$. The result follows from the chain of inclusions

$$(H_1 \cap H_2)^\circ \subset H_1^\circ \cap H_2^\circ \subset H_1^\circ \cap H_2 \subset H_1 \cap H_2. \quad \square$$

Proposition 2.16. *Conjecture 2.14 implies Conjecture 2.3.*

Proof. Since Z is integral, it is contained in one translate $\tau + \langle Z \rangle^\circ$ of the identity component $\langle Z \rangle^\circ$ of the group it generates, for some (possibly trivial) torsion point τ . Hence the translate $-\tau + Z$ is contained in $\langle Z \rangle^\circ$.

We will apply Conjecture 2.14 to the abelian variety $\langle Z \rangle^\circ$ and subvariety $V := -\tau + Z$. Let $H_e := \langle Z \rangle^\circ \cap \langle -\tau + A^r[\mathbf{e}] \rangle$. Applying Lemma 2.15 twice, using the fact that τ is a torsion point, we have

$$\dim \langle Z \rangle \cap A^r[\mathbf{e}] = \dim \langle Z \rangle^\circ \cap A^r[\mathbf{e}] = \dim \langle Z \rangle^\circ \cap \langle -\tau + A^r[\mathbf{e}] \rangle = \dim H_e.$$

Since $\dim \langle Z \rangle^\circ = \dim \langle Z \rangle$ and $\dim V = \dim Z$, we have

$$\mathcal{E}_Z = \{\mathbf{e} : \dim \langle Z \rangle \cap A^r[\mathbf{e}] \geq \dim \langle Z \rangle - \dim Z\} = \{\mathbf{e} : \dim H_e \geq \dim \langle Z \rangle^\circ - \dim V\}.$$

Thus, if $\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z$, then $\dim H_e + \dim V < \dim \langle Z \rangle^\circ$. Hence Conjecture 2.14 implies that

$$V \cap \left(\bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} H_e \right)$$

is not Zariski dense in V .

Considering translation, we have

$$-\tau + (Z \cap A^r[\mathbf{e}]) = (-\tau + Z) \cap (-\tau + A^r[\mathbf{e}]) \subset (-\tau + Z) \cap \langle -\tau + A^r[\mathbf{e}] \rangle = V \cap H_e,$$

and hence

$$Z \cap \left(\bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} A^r[\mathbf{e}] \right) \subset \tau + \left(V \cap \left(\bigcup_{\mathbf{e} \in \mathbb{Z}^r \setminus \mathcal{E}_Z} H_e \right) \right)$$

is not Zariski dense in $\tau + V = Z$, as desired. $\square$

3. RELATING UNLIKELY SPLITTING PROFILES TO UNLIKELY INTERSECTIONS

The goal of this section is to show how the unlikely intersections results from Section 2 imply the results on unlikely splitting profiles from the introduction. We first give the general framework and then in Section 3.2, we explain how to deduce the specific statements in the introduction.

3.1. The general framework. The main focus of this paper is determining for which places v a given splitting profile $\mathbf{P}$ is realized by points $x \in C$ with $[x] \in \langle [D] \rangle + H$, for some k -rational divisor D and abelian subvariety $H \subset J_C$. The applications will all rely on the case when $H = \{0\}$, but we give the framework for arbitrary H with a view to future exploration of unlikely splitting profiles in the case when J_C is positive rank.

For v a nonarchimedean place of good reduction, we can spread out and make use of reduction modulo v . The first step in our approach is to observe that existence of *global* points with prescribed ramification at v implies the existence $\mathbb{F}_v$ -points in a certain subvariety of a product of symmetric products.

Lemma 3.1. *Let C be a nice curve over a number field k , let v be a prime of good reduction, let D be a k -rational divisor of positive degree on C , let $x \in C$ be a closed point, and let $H \subset J_C$ be an abelian subvariety. Write $\mathbf{P} := \mathbf{P}_v(\mathbf{k}(x)/k)$, $r := r(\mathbf{P})$, $\mathbf{f} := \mathbf{f}(\mathbf{P})$, $\mathbf{e} := \mathbf{e}(\mathbf{P})$. If $[x] \in \langle [D] \rangle + H$, then there exist points $\bar{x}_1, \dots, \bar{x}_r$ on $C_{\mathbb{F}_v}$ with $\deg(\bar{x}_i) \mid f_i$ such that*

$$\sum_i e_i \cdot \frac{f_i}{\deg(\bar{x}_i)} [\bar{x}_i] \in \frac{\deg(x)}{\deg(D)} [D_{\mathbb{F}_v}] + H_{\mathbb{F}_v}.$$

In particular, there is an $\mathbb{F}_v$ -point of $\text{Sym}_{C_{\mathbb{F}_v}}^{\mathbf{f}}$ that maps to $\frac{\deg(x)}{\deg(D)} [D_{\mathbb{F}_v}] + H_{\mathbb{F}_v}$ under the composition

$$\text{Sym}_C^{\mathbf{f}} \xrightarrow{\text{Abel-Jacobi}} W_C^{\mathbf{f}} \xrightarrow{[\mathbf{e}]} \text{Pic}_C^{\deg(\mathbf{P})}. \quad (3.1)$$

Further, given an integral model $\mathcal{C} \rightarrow \text{Spec } \mathcal{O}_k$, if the map $\text{Spec } \mathcal{O}_{\mathbf{k}(x)} \rightarrow \mathcal{C}$, corresponding to the point x is injective over $v \in \text{Spec } \mathcal{O}_k$ (which occurs for all but finitely many primes), then $\deg(\bar{x}_i) = f_i$ and $\bar{x}_i \neq \bar{x}_j$ for $i \neq j$.

Proof. Let $\mathcal{C} \rightarrow \text{Spec } \mathcal{O}_{k,S}$ be a smooth proper model of C/k . The closed point x spreads out to a map $\text{Spec } \mathcal{O}_{\mathbf{k}(x),S'} \rightarrow \mathcal{C}$, where S' is the set of places of $\mathbf{k}(x)$ above S . By definition of the splitting profile, the fiber of $\text{Spec } \mathcal{O}_{\mathbf{k}(x),S'}$ over $v \in \text{Spec } \mathcal{O}_{k,S}$ is r distinct points of degree f_i over $\text{Spec } \mathbb{F}_v$ and multiplicity e_i . The image of each of these points is a closed point $\bar{x}_i \in C_{\mathbb{F}_v}$ of degree dividing f_i . Hence, the fiber of the image of $\text{Spec } \mathcal{O}_{\mathbf{k}(x),S'} \rightarrow \mathcal{C}$ over v is equal (as a divisor) to $\sum_i e_i \frac{f_i}{\deg(\bar{x}_i)} \bar{x}_i$. If the map $\text{Spec } \mathcal{O}_{\mathbf{k}(x),S'} \rightarrow \mathcal{C}$ is injective at v , then $\deg(\bar{x}_i) = f_i$ and $\bar{x}_i \neq \bar{x}_j$ for $i \neq j$.

Since $[x] \in \langle [D] \rangle + H$ is a closed condition, it also holds modulo v . In other words, the $\mathbb{F}_v$ -point $\left(\frac{f_i}{\deg(\bar{x}_i)} \bar{x}_i \right)_i \in \text{Sym}_{C_{\mathbb{F}_v}}^{\mathbf{f}}$ maps to $\frac{\deg(x)}{\deg(D)} [D_{\mathbb{F}_v}] + H_{\mathbb{F}_v}$. $\square$

Since the morphism (3.1) and the subvariety $\frac{\deg(x)}{\deg(D)} [D] + H$ are defined over k , this gives a geometric criterion to determine whether a fixed splitting profile $\mathbf{P}$ is likely in $|\frac{\deg(\mathbf{P})}{\deg(D)} D + H|$.

Lemma 3.2. *Let C be a nice curve over a number field k , let D be a k -rational divisor of positive degree on C , and let $\mathbf{P}$ be a splitting profile with $\deg(D) \mid \deg(\mathbf{P})$. If the image of the composition*

$$\text{Sym}_C^{\mathbf{f}} \xrightarrow{\text{Abel-Jacobi}} W_C^{\mathbf{f}} \xrightarrow{[\mathbf{e}]} \text{Pic}_C^{\deg(\mathbf{P})},$$

does not intersect $|\frac{\deg(\mathbf{P})}{\deg(D)} D| + H$, then the splitting profile $\mathbf{P}$ is unlikely in $|\frac{\deg(\mathbf{P})}{\deg(D)} D + H|$.

Proof. For any finite set of places S and variety $\mathcal{V} \rightarrow \text{Spec } \mathcal{O}_{k,S}$, the set of places $v \notin S$ where $\mathcal{V}$ has an $\mathbb{F}_v$ -point is a subset of the support $\text{Supp}(\mathcal{V}) \subset \text{Spec } \mathcal{O}_{k,S}$. In particular, if $\mathcal{V}_k = \emptyset$, there are finitely many places v where $\mathcal{V}$ has an $\mathbb{F}_v$ -point. Now we apply this for $\mathcal{V}$ an

integral model of $|\frac{\deg(\mathbf{P})}{\deg(D)}D + H| \subset \text{Sym}_C^{\deg(\mathbf{P})}$, which is the complete preimage of $\frac{\deg(\mathbf{P})}{\deg(D)}[D] + H$ under the composition of maps (3.1). $\square$

The geometric criterion in Lemma 3.2 is most useful when one is interested in a finite set of possible profiles, each of manageable degree. Indeed, Lemmas 3.1 and 3.2 give us a method for computing the sets of primes realizing a given splitting profile, such as is given in Table 1.1; see Section 3.2.3 for more details. However, since the intersection depends on all aspects of the splitting profile, it is of limited use when considering an infinite set of profiles as discussed in the main theorems of the paper.

One of the main insights in the paper is that for the (infinitely many) splitting profiles that share a common $\mathbf{f}$ tuple, the intersections under study can be placed in the common framework of the intersection of a fixed subvariety of $(J_C/H)^r$ with a (varying) algebraic subgroup. When $\mathbf{f} = (f)$ has length 1, we intersect with the torsion subgroups of J_C/H . In general, we intersect with the subgroups $A^r[\mathbf{e}(\mathbf{P})]$ studied in Section 2, where $A = J_C/H$.

Proposition 3.3. *Let C be a nice curve over a number field k , let D be a k -rational divisor of positive degree on C , and let $H \subset J_C$ be an abelian subvariety. Let $\mathbf{P}^0 = \{(e_i^0, f_i)\}$ be splitting profile with $\deg(D) \mid e_i^0 f_i$ for all i and write $\mathbf{e}^0 := \mathbf{e}(\mathbf{P}^0)$ and $\mathbf{f} := \mathbf{f}(\mathbf{P}^0)$. For $\mathbf{e} = (e_1, \dots, e_r) \in \text{Diag}(\mathbf{e}^0)\mathbb{Z}_{>0}^r$, if the intersection of k -varieties*

$$(\pi_H^r \circ \iota_D \circ [\text{Diag}(\mathbf{e}^0)]) (W_C^{\mathbf{f}}) \cap (J_C/H)^r[\mathbf{e}/\mathbf{e}^0] \quad (3.2)$$

is empty, then $\mathbf{P} := \{(e_i, f_i)\}$ is unlikely in $|\frac{\deg(\mathbf{P})}{\deg(D)}D + H|$.

Remark 3.4. Any splitting profile $\mathbf{P}$ for which $\deg(D) \mid ef$ for all $(e, f) \in \mathbf{P}$ can be understood in the context of Proposition 3.3 with

$$\mathbf{P}^0 := \left\{ \left(\frac{\deg(D)}{\gcd(f, \deg(D))}, f \right) : (e, f) \in \mathbf{P} \right\}.$$

Proof. By definition of the subgroup $(J_C/H)^r[\mathbf{e}/\mathbf{e}^0]$, the intersection (3.2) is empty if and only if the image of the composition

$$W_C^{\mathbf{f}} \xrightarrow{[\text{Diag}(\mathbf{e}^0)]} \prod_i W_C^{e_i^0 f_i} \xrightarrow{\iota_D} J_C^r \xrightarrow{\pi_H^r} (J_C/H)^r \xrightarrow{[\mathbf{e}/\mathbf{e}^0]} J_C/H \quad (3.3)$$

does not contain 0. This composition fits into the following commutative diagram.

$$\begin{array}{ccccccc} \text{Sym}_C^{\mathbf{f}} & \longrightarrow & W_C^{\mathbf{f}} & \xrightarrow{[\text{Diag}(\mathbf{e}^0)]} & \prod_i W_C^{e_i^0 f_i} & \xrightarrow{\iota_D} & J_C^r \xrightarrow{\pi_H^r} (J_C/H)^r \\ & & \searrow [\mathbf{e}] & & \downarrow [\mathbf{e}/\mathbf{e}^0] & & \downarrow [\mathbf{e}/\mathbf{e}^0] \\ & & & & \text{Pic}_C^{\deg(\mathbf{P})} & \xrightarrow{\iota_D} & J_C \xrightarrow{\pi_H} J_C/H \end{array}$$

Since along the bottom row, the complete preimage of 0 is $[\frac{\deg(\mathbf{P})}{\deg(D)}D] + H$, the image of the composition (3.3) does not contain 0 if and only if the image of (3.1) does not meet $[\frac{\deg(\mathbf{P})}{\deg(D)}D] + H$. Thus, the assumptions of the proposition allow us to conclude by applying Lemma 3.2. $\square$

While Proposition 3.3, may not appear to be substantially different than Lemma 3.2, the fact that one component of the intersection (3.2) depends only on $\mathbf{P}^0$ and the other component

is an algebraic subgroup means that the union over $\mathbf{e} \in [\text{Diag}(\mathbf{e}^0)]\mathbb{Z}^r$ of intersections (3.2) falls under the framework of unlikely intersections (hence our choice of terminology).

When $r = 1$, we prove that there are a positive density set of degrees d where total ramification is unlikely in the linear system $|dD|$ for D an ineffective divisor on a curve of genus at least 2. More generally, we use the Manin–Mumford conjecture to prove the following.

Theorem 3.5. *Let C be a nice curve over a number field k , let $D \in \text{Pic}_C(k)$ be a divisor of positive degree, and let $H \subset J_C$ be an abelian subvariety. Let $\mathbf{P}^0 = \{(e_i^0, f_i)\}$ be a splitting profile such that $\deg(D) \mid e_i^0 f_i$ for all i and write $\mathbf{e}^0 := \mathbf{e}(\mathbf{P}^0)$ and $\mathbf{f} := \mathbf{f}(\mathbf{P}^0)$.*

If $[\mathbf{e}^0](W_C^{\mathbf{f}}) \cap \left(\frac{\deg(\mathbf{P}^0)}{\deg(D)}[D] + H\right) = \emptyset$, then there exists an $m = m(C, \mathbf{P}^0) \in \mathbb{Z}_{>0}$ such that

$$(\pi_H \circ \iota_D \circ [\mathbf{e}^0])(W_C^{\mathbf{f}}) \cap J_C/H[d] = \emptyset \quad \text{for all } d \text{ with } \gcd(d, m) = 1.$$

Moreover, if m has the above property, then for all d coprime to m , the splitting profile $\mathbf{P} = \{(de_i^0, f_i)_{i=1}^r\}$ is unlikely in $\left|\frac{\deg(\mathbf{P})}{\deg(D)}[D] + H\right|$.

Remark 3.6. As we will see in the proof, the first statement about m is a straightforward consequence of the Manin–Mumford conjecture (Proposition 2.1). The novelty is in the last statement.

Proof of Theorem 3.5. Since we assumed that $[\mathbf{e}^0](W_C^{\mathbf{f}}) \cap \left(\frac{\deg(\mathbf{P}^0)}{\deg(D)}[D] + H\right) = \emptyset$, we have

$$0 \notin Z := (\pi_H \circ \iota_D \circ [\mathbf{e}^0])(W_C^{\mathbf{f}}) \subset J_C/H.$$

Since $0 \notin Z$, Proposition 2.1 implies that there exists a finite list of integers $m_1, \dots, m_s \in \mathbb{Z}_{\geq 2}$ such that

$$Z \cap (J_C/H)[d] = \emptyset \quad \Longleftrightarrow \quad d \notin \bigcup_i m_i \mathbb{Z}.$$

In particular, the first statement in Theorem 3.5 holds for any positive integer m divisible by $\text{lcm}(m_1, \dots, m_s)$.

Now take such an m , and a positive integer d coprime to m , which implies that $Z \cap (J_C/H)[d] = \emptyset$. Let $\mathbf{e} = \mathbf{e}(\mathbf{P}) = (de_i^0)_{i=1}^r$. Since $\sigma \circ \iota_D \circ [\text{Diag}(\mathbf{e}^0)] = \iota_D \circ [\mathbf{e}^0]$, the composition of π_H with the summation map σ sends $(\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W_C^{\mathbf{f}}) \cap J_C^r[\mathbf{e}/\mathbf{e}^0]$ to $Z \cap (J_C/H)[d]$ and so

$$(\pi_H^r \circ \iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W_C^{\mathbf{f}}) \cap J_C/H^r[\mathbf{e}/\mathbf{e}^0] = \emptyset.$$

Thus, by Proposition 3.3, the splitting profile $\mathbf{P}$ is unlikely in $\left|\frac{\deg(\mathbf{P})}{\deg D}D + H\right|$. $\square$

Theorem 3.5 gives a specific criterion, depending on a single splitting profile $\mathbf{P}^0$, to determine when unlikely splitting profiles arise when we allow the ramification degrees to vary over the rank 1 subgroup of $\mathbb{Z}^r$ generated by $\mathbf{e}(\mathbf{P}^0)$. Restricting to a rank 1 subgroup of ramification degrees is a crucial assumption that allows us to leverage the Manin–Mumford conjecture. The more general problem of studying the variation over high rank subgroups of $\mathbb{Z}^r$ relates to intersections that arise in the (still open) Zilber–Pink conjecture. Conditional on the Zilber–Pink conjecture (or rather a special case given in Conjecture 2.3), we obtain the following generalization of Theorem 3.5 to arbitrary splitting profiles.

Theorem 3.7. *Let C be a nice curve over a number field k , let $D \in \text{Pic}_C(k)$ be a divisor of positive degree and let $H \subset J_C$ be an abelian subvariety. Let $\mathbf{P}^0 = \{(e_i^0, f_i)\}$ be a splitting profile such that $\deg(D) | e_i^0 f_i$ for all i .*

Assume Conjecture 2.3 and assume that there exists an $\tilde{\mathbf{e}} \in \mathbb{Z}^r$ such that for every abelian subvariety $B \subset J_C/H$ and every torsion point $\tau \in (J_C/H)^r$, either

- $\dim(\pi_H^r \circ \iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) \cap (\tau + B^r) < \dim B$; or
- $[\tilde{\mathbf{e}}]\tau \notin B$.

Then there exists a finite set of subgroups $M_1, \dots, M_s \subsetneq \mathbb{Z}^r$ with $\tilde{\mathbf{e}} \notin M_i$ for every subgroup M_i of finite index (which implies that $\bigcup_i M_i \subsetneq \mathbb{Z}^r$) such that for each $\mathbf{e} \in \text{Diag}(\mathbf{e}^0)(\mathbb{Z}^r \setminus \bigcup M_i)$, the splitting profile $\mathbf{P} = \{(e_i, f_i)_{i=1}^r\}$ is unlikely in $\left\lfloor \frac{\deg(\mathbf{P})}{\deg(D)} D + H \right\rfloor$.

Proof. Applying Theorem 2.5 to

$$Z := (\pi_H^r \circ \iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) \subset (J_C/H)^r,$$

there exist finitely many proper subgroups $M_1, \dots, M_r \subset \mathbb{Z}^r$ with $\bigcup_{i=1}^s M_i \neq \mathbb{Z}^r$ such that $Z \cap \bigcup_{\mathbf{e} \in \mathbf{e}^0 \notin \bigcup M_i} (J_C/H)^r[\mathbf{e}/\mathbf{e}^0] = \emptyset$. The conclusion follows from Proposition 3.3. $\square$

3.2. Proofs of Theorems in the introduction. Theorems 1.1 and 1.6 are obtained as corollaries of Theorem 3.5 (Section 3.2.1). Theorem 1.8 is obtained as a corollary of Theorem 3.7 (Section 3.2.2).

3.2.1. Corollaries of Theorem 3.5.

Proof of Theorem 1.1. Since the Jacobian of C has rank 0, there are finitely many divisor classes of a given positive degree. Set $w := \text{index}(C)$ and let $D_1, \dots, D_n$ be representatives for the complete list of k -rational divisor classes of degree w . (By definition of the index, one of these divisor classes has a k -rational representative, but others may not.)

We claim that $[w](C) \subset \text{Pic}_C^w$ does not contain $[D_i]$ for any i . Since $[D_i]$ is k -rational and the morphism $[w]$ is defined over k , if there is a geometric point $x \in C(\bar{k})$ such that $w[x] = [D_i]$, the same is true for all Galois conjugates of x . By assumption, $C(k) = \emptyset$, so there exists a conjugate $y \in C(\bar{k})$, different from x , such that $w[x] = w[y] = [D_i]$. In particular, $|D_i|$ is a basepointfree linear system and there is a degree w map from C to a curve of genus 0 with a totally ramified point. However, such a map was excluded by hypothesis.

Thus we may apply Theorem 3.5 with $D = D_i, r = 1, \mathbf{P}^0 = \{(w, 1)\}$ and $H = 0$. Then we obtain a positive integer m_i such that total ramification is unlikely in $|dD_i|$ for all d coprime to m_i .

Let $m = \text{lcm}(m_1, \dots, m_n, \#J_C(k))$. Then for every d coprime to m , the degree dw points are contained in $\bigcup_{j=1}^n |dD_j|$, and in each linear system total ramification is unlikely. Hence total ramification is unlikely for all degree dw points. $\square$

Proof of Theorem 1.6. Since the Jacobian of C has rank 0, there are finitely many divisor classes of a given positive degree. Set $w := \text{index}(C)$ and let $D_1, \dots, D_n$ be representatives for the complete list of k -rational divisor classes of degree w .

Since $\text{index}(C) | e_i^0 f_i$ by assumption, we have, $\deg(D_j) | e_i^0 f_i$ for all j and i . Further, since there are no points of degree at most $\deg(\mathbf{P}^0)$, then $\left\lfloor \frac{\deg(\mathbf{P}^0)}{\deg(D_j)} D_j \right\rfloor \notin W_C^{\deg(\mathbf{P}^0)}$ (and hence not in $[\mathbf{e}^0]W_C^{\mathbf{f}} \subset W_C^{\deg(\mathbf{P}^0)}$). Thus we may apply Theorem 3.5 with $D = D_j, H = 0$ to obtain a

positive integer m_j such that the splitting profile $P = \{(de_i^0, f_i)_{i=1}^r\}$ is unlikely in $|d \frac{\deg(P^0)}{\deg(D_j)} D_j|$ for all d coprime to m_j .

Let $m := \text{lcm}(m_1, \dots, m_n, \#J_C(k))$. Then if d is coprime to m , it is coprime to m_j for all j and hence the splitting profile $P = \{(de_i^0, f_i)_{i=1}^r\}$ is unlikely in $|\frac{\deg(P)}{\deg(D_j)} D_j|$ for all j . Further, since d and $\frac{\deg(P^0)}{\deg(D_j)}$ are coprime to $\#J_C(k)$, every point of degree $\deg(P)$ is linearly equivalent to $\frac{d \deg(P^0)}{\deg(D_j)} D_j$ for some j . Hence, the splitting profile P is unlikely in C . $\square$

The careful reader will note that the proofs of Theorem 1.6 and 1.1 are very similar. Indeed, Theorem 1.1 could almost be a corollary of Theorem 1.6 by taking $P^0 = \{(\text{index}(C), 1)\}$ except that we have weakened the hypothesis that there are no points of degree at most $\deg(P^0) = \text{index}(C)$ by instead assuming that $C(k) = \emptyset$ and that for all maps $f: C \rightarrow Z$ with $\deg(f) = \text{index}(C)$ and Z a curve of genus 0 there are no points of total ramification. This is the minimal hypothesis necessary for Theorem 1.1 (see Corollary 4.9 for the converse).

3.2.2. Corollaries of Theorem 3.7.

Proof of Theorem 1.8. Since the Jacobian of C has rank 0, there are finitely many divisor classes of a given positive degree. Set $w := \text{index}(C)$ and let $D_1, \dots, D_n$ be representatives for the complete list of k -rational divisor classes of degree w . By assumption, $\deg(D_j) | e_i^0 f_i$ for all j and i . Without loss of generality, assume that $e_1^0 f_1 = \min_i (e_i^0 f_i)$. Since we assumed that C has no points of degree $\min(e_i^0 f_i) = e_1^0 f_1$, we know that $\frac{e_1^0 f_1}{\deg(D_j)} D_j$ is ineffective for all j . Hence, for each j , the locus

$$(\iota_{D_j} \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) \subset J_C^r$$

does not intersect $\text{pr}_1^{-1}(0)$, where $\text{pr}_1: J_C^r \rightarrow J_C$ denotes projection onto the first factor.

For each prime divisor $p_i | \#J_C(k)$, let $M_i^0 \subset \mathbb{Z}^r$ be the finite index subgroup of $\mathbb{Z}^r$ consisting of tuples $(x_1, \dots, x_r)$ such that

$$\sum_{\ell=1}^r \left(\frac{e_{\ell}^0 f_{\ell}}{\text{index}(C)} \right) x_{\ell} \equiv 0 \pmod{p_i}.$$

If $\mathbf{e}/\mathbf{e}^0 \in \mathbb{Z}^r \setminus \cup_i M_i^0$, then $\deg(P) = \sum_{\ell=1}^r \left(\frac{e_{\ell} f_{\ell}}{\text{index}(C)} \right)$ is relatively prime to $\#J_C(k)$ and so every point of degree $\deg(P)$ is linearly equivalent to $\frac{\deg(P)}{\deg(D_j)} D_j$ for some j . By our assumption that $e_1^0 f_1 / \text{index}(C)$ is relatively prime to $\#J_C(k)$, we know that the M_i^0 does not contain the first standard basis vector for any i .

We claim that we can apply Theorem 3.7 for each $D = D_j$ with $H = 0$ and $\tilde{\mathbf{e}} = (1, 0, \dots, 0)$ the first standard basis vector. Since J_C is geometrically simple by assumption, the only abelian subvarieties $B \subset J_C$ that we need to consider are $B = J_C$ and $B = 0$. For $B = J_C$, we have $\tau + B^r = B^r$ for every torsion point τ , and

$$\dim(\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) \cap B^r = \dim(\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) = \sum_{i=1}^r f_i < g = \dim B$$

by assumption. For $B = 0$ and $\tau \in J_C^r$ a torsion point, we have $\dim((\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}}) \cap (\tau + B^r)) \geq \dim B = 0$ if and only if $\tau \in (\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}})$. Thus, we must show that whenever $\tau \in (\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}})$, then $[\tilde{\mathbf{e}}]\tau \notin B = 0$. This follows since if $\tau \in (\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}})$, then $[\tilde{\mathbf{e}}](\tau) = \text{pr}_1(\tau) \in \text{pr}_1(\iota_D \circ [\text{Diag}(\mathbf{e}^0)])(W^{\mathbf{f}})$, and, as explained in the first paragraph,

does not contain 0 by our assumptions. Hence, for each j , Theorem 3.7 with $D = D_j$ and $H = 0$ implies the existence of subgroups $M_1^j, \dots, M_{s_j}^j \subset \mathbb{Z}^r$ such that $\tilde{\mathbf{e}} \notin M_i^j$ for each rank r subgroup M_i^j and for all $\mathbf{e} \in \text{Diag}(\mathbf{e}^0)(\mathbb{Z}^r \setminus \bigcup M_i^j)$, the splitting profile $\mathbf{P} = \{(e_i, f_i)\}$ is unlikely in $\left\lfloor \frac{\deg \mathbf{P}}{\deg D_j} D_j \right\rfloor$.

Consider the following finite unions of subgroups:

$$M := \bigcup_{j=0}^n \bigcup_{i=1}^{s_j} M_i^j \quad \text{and} \quad \tilde{M} := \bigcup_{j=0}^n \bigcup_{\substack{i=1 \\ \text{rank}(M_i^j)=r}}^{s_j} M_i^j.$$

By construction, $\tilde{M} \not\ni \tilde{\mathbf{e}}$, and so $\tilde{M}$ cannot equal all of $\mathbb{Z}^r$. Since $\mathbb{Z}^r \setminus \tilde{M}$ must contain a coset of a finite index subgroup [VV26, Lemma 4.3] and M is the union of $\tilde{M}$ and finitely many subgroups of rank strictly less than r , the finite union M also cannot equal all of $\mathbb{Z}^r$ (even though it may contain $\tilde{\mathbf{e}}$). We claim that the theorem holds with this proper finite union of subgroups M . Indeed, if $\mathbf{e} \in \text{Diag}(\mathbf{e}^0)(\mathbb{Z}^r \setminus M)$, then every point of degree $\deg(\mathbf{P})$ is in $\left\lfloor \frac{\deg \mathbf{P}}{\deg D_j} D_j \right\rfloor$ for some j , so the result follows by Theorem 3.7. $\square$

3.2.3. Proof of Table 1.1. In this Section we show how Theorem 3.5 yields the following explicit computational result:

Proposition 3.8. *Let C be a curve from Table 1.1. Then for every odd d , the set of places*

$$\mathcal{T}_d(C) := \{p : \exists x \in C, \deg(x) = d, \mathbf{k}(x)/\mathbb{Q} \text{ totally ramified at } p\}$$

is finite. Moreover, for $d = 3, 5$, the sets $\mathcal{T}_d(C) \cap \{p : p \nmid 2 \text{cond}(C)\}$ are as listed in column 2 and 3 of Table 1.1.

Proof. From the L-functions and Modular Forms Database (LMFDB) and Poonen's code to compute torsion packets [LMFDB, Poo01, PARI/GP], each curve listed in Table 1.1 has the following properties (see below for more details):

- (1) the Weierstrass points of C split into two degree 3 points, w, w' (which necessarily implies $[w] = [w']$ and $2[w] = 3K_C$),
- (2) the $\mathbb{Q}$ -points of the Picard scheme are $\{nK_C, nK_C - [w] : n \in \mathbb{Z}\}$,
- (3) for any geometric Weierstrass point $p \in C(\bar{k})$, the intersection $(C - p) \cap (J_C)_{\text{tors}}$ consists only of the Weierstrass points.

Indeed, condition (1) can be verified by taking a simple Weierstrass model $y^2 = f(x)$ for C (which is given in the LMFDB) and observing that over $\mathbb{Q}$, the polynomial $f(x)$ factors as a product of two irreducible cubics. Then, given (1), which in particular implies that C has index 1, Condition (2) is equivalent to the condition that $J_C(\mathbb{Q}) = \{O\}$, which is computed in the LMFDB. Finally, [Poo01] gives an algorithm to compute $(C - p) \cap (J_C)_{\text{tors}}$, and we use Poonen's PARI/GP implementation to verify that this set of geometric points has order 6. Since this set contains every Weierstrass point (the difference of any two Weierstrass points is 2-torsion), this proves that no other points are contained in this intersection.

Now we will use these properties together with Theorem 3.5 to prove the proposition. Condition (2) says that there is a unique divisor class in every degree, and they are all integer multiples of $2K_C - [w]$. In addition, $2K_C - [w] = [w] - K_C$ is not effective and so we may apply Theorem 3.5 with $D := 2K_C - [w]$, $\mathbf{P}^0 := \{(1, 1)\}$, and $H = 0$. We claim that $\iota_D(C) \cap J_C[d] = \emptyset$ for all odd d , which implies that in Theorem 3.5 we may take $m = 2$. Thus

we obtain the desired conclusion about finiteness. We will prove this claim and then explain how we compute the sets $\mathcal{T}_d(C) \cap \{p : p \nmid 2\text{cond}(C)\}$

Let $q \in C(\bar{k})$ be such that $[q] - [D] \in (J_C)_{\text{tors}}$. Then $2[q] - K_C$ is torsion, so for any geometric Weierstrass point $p \in C(\bar{k})$, the difference $[q] - [p]$ is torsion. By Condition (3), q must be a geometric Weierstrass point, so $2([q] - D) = K_C - K_C = 0$. Since D is ineffective, $[q] \neq D$ and so $\iota_D(C) \cap J_C[d] = \emptyset$ for all odd d .

Now we explain how to compute the sets $\mathcal{T}_d(C) \cap \{p : p \nmid 2\text{cond}(C)\}$. Recall that condition (1) implies that C has a model of the form $y^2 = g_0(x)g_1(x)$ for irreducible cubics g_0, g_1 . Moreover, the polynomials g_0, g_1 are separable and relatively prime for every odd prime not dividing the conductor of C . The complete linear system $|3D| = |w|$ corresponds to the morphism sending a point $z \in C$ to $[g_0(z) : y(z)] = [y(z) : g_1(z)]$. Thus if the fiber above a point $[t_0 : t_1] \in \mathbb{P}^1$ give a point that is totally ramified at v then the cubic polynomial $t_1^2 g_0 - t_0^2 g_1$ is a scalar multiple of a perfect cube modulo v . One can compute the locus of such v using elimination ideals (see the Magma code in the [Github repository](#) for this paper). Further, for each of the v that appear, we verify that this value of $[t_0 : t_1]$ can be lifted such that $t_1^2 g_0 - t_0^2 g_1$ is irreducible over k_v and defines a totally ramified field extension.

For $d = 5$, we study the linear system $|5D|$, which is very ample. Since $h^0(5D) = h^0(2D)h^0(3D)$, the morphism corresponding to $|5D|$ factors as the composition

$$C \xrightarrow{(\varphi_{|2D|}, \varphi_{|3D|})} \mathbb{P}^1 \times \mathbb{P}^1 \xrightarrow{(\varphi_{(1,1)})} \mathbb{P}^3$$

Thus, a hyperplane section corresponds to intersecting C (viewed as a $(2, 3)$ curve) with a $(1, 1)$ curve. We wish to apply Theorem 4.1, so we must find a hyperplane section modulo v that meets C at an $\mathbb{F}_v$ -point with multiplicity 5. Such a hyperplane section necessarily corresponds to an integral $(1, 1)$ curve $Z \simeq \mathbb{P}^1$. Thus, we may compute the intersection of C with Z as the vanishing of a single variable degree 5 polynomial on Z . By Lemma 3.2, a place of good reduction $v \in \mathcal{T}_5(C)$ must give rise to a Z such that the associated degree 5 polynomial is a scalar multiple of a fifth power. As above, this can be computed using elimination ideals (see the Magma code in the [Github repository](#) for this paper). For each of the v that appear, we could verify that this value of $[t_0 : t_1]$ can be lifted such that the degree 5 polynomial is irreducible over k_v and defines a totally ramified field extension, as we did in the degree 3 case. Alternately, since $|5D|$ is very ample, we can appeal to our lifting result (Theorem 4.1) that we prove in the next section. $\square$

Given a genus 2 curve which has properties (1) and (2), but not (3), then there will be some odd prime p for which $\mathcal{T}_p(C)$ is infinite.

Example 3.9. Let C be the genus 2 curve with LMFDB label [52969.b.52969.1](#). This curve has a geometric point whose difference with a geometric Weierstrass point is 3-torsion in the Jacobian, and thus the set $\mathcal{T}_3(C)$ is infinite. One can compute this set using elimination theory as described above, and in doing so, we determine that the set $\mathcal{T}_3(C) \cap \{p : p \nmid 2 \cdot 52969\}$ consists of all primes different from 2 and 23 such that $-7 \in \mathbb{F}_p^{\times 2}$, i.e., all primes different from 2 and 23 that are congruent to 1, 2, or 4 modulo 7.

The methods of this paper do not directly shed light on the sets $\mathcal{T}_d(C)$ themselves (beyond finiteness). For example, to the best of our knowledge, the following questions remain open.

- (1) Can the sets $\mathcal{T}_d(C)$ be arbitrary large? If so, then for d in a fixed congruence class modulo $m(C)$, what are the asymptotics of $\#\mathcal{T}_d(C)$?

- (2) What is the Dirichlet density of $\cup_{(d,m(C))=1} \mathcal{T}_d(C)$? We can computationally verify (see code in the [Github repository](#) for this paper) that for the curve C with LMFDB label [7676.a.7676.1](#), the prime 31 is not in $\mathcal{T}_d(C)$ for any odd integer d (meaning there are no points on C of odd degree where 31 is totally ramified in the residue field). In particular, this example shows that $\cup_{(d,m(C))=1} \mathcal{T}_d(C)$ can be a proper subset of all primes of good reduction. However, beyond this, little is known.

4. REALIZING SPLITTING PROFILES

In this section, we explore the complementary problem of showing that certain splitting profiles can be realized. One of the main techniques in this section is a lifting result: that given an effective representative D of a sufficiently positive (globally defined) divisor class over a finite field $\mathbb{F}$, the representative can be lifted to a point over a global field in the same linear system with local behavior prescribed by the divisor D . Precisely, we prove the following.

Theorem 4.1. *Let C/k be a nice curve, let v be a place of k such that C has good reduction at v and $\mathbb{F}_v$ is sufficiently large, let D be a divisor on C of positive degree, let $\mathbf{P}$ be a splitting profile with $\deg(\mathbf{P}) = \deg(D)$ and let $\mathbf{e} = \mathbf{e}(\mathbf{P}), \mathbf{f} = \mathbf{f}(\mathbf{P})$ be the associated tuples. Assume that $h^0(C, D) = h^0(C_{\mathbb{F}_v}, D_{\mathbb{F}_v})$ and $|D_{\mathbb{F}_v}|$ is very ample. If there exist distinct points $\bar{x}_1, \dots, \bar{x}_r(\mathbf{P}) \in C_{\mathbb{F}_v}$ with $(\deg(\bar{x}_i))_i = \mathbf{f}$ such that*

$$\sum_i e_i \bar{x}_i \sim D_{\mathbb{F}_v}. \quad (4.1)$$

then there exists a degree d point $x \in C$ with splitting profile $\mathbf{P}$ at v and with $x \sim D$.

This is morally a Hensel's Lemma and Hilbert irreducibility argument, although the proof requires care because we are lifting a (possibly) nonsmooth scheme. In Section 4.1, we prove the nonarchimedean lifting result and in Section 4.2, we explain how to deduce Theorem 4.1 from the local result. This lifting result allows us to give many examples of likely splitting profiles; these examples are collected in Section 4.3, along with a proof of Proposition 1.10.

4.1. Lifting effective divisors p -adically. Let K be a p -adic field with valuation ring $\mathcal{O} = \mathcal{O}_K$ and residue field $\mathbb{F} = \mathbb{F}_K$. Given a smooth $\mathcal{O}$ -scheme $\mathcal{C} \hookrightarrow \mathbb{P}_{\mathcal{O}}^N$ of relative dimension 1, the goal is to lift a linear section $V(\bar{\ell}) \subset \mathbb{P}_{\mathbb{F}}^N$ that meets $\mathcal{C}_{\mathbb{F}}$ possibly with multiplicity, to a linear section over $\mathcal{O}$ that meets $\mathcal{C}$ with multiplicity 1 at every point in its support. Precisely, we prove the following

Proposition 4.2. *Let $\mathcal{C} \hookrightarrow \mathbb{P}_{\mathcal{O}}^N$ be a smooth $\mathcal{O}$ -scheme of relative dimension 1, let $\mathbf{P}$ be a splitting profile and let $\mathbf{e} = \mathbf{e}(\mathbf{P}), \mathbf{f} = \mathbf{f}(\mathbf{P})$ be the associated tuples. Assume that $r(\mathbf{P}) < \#\mathbb{F}$ and that there exists a linear form $\bar{\ell} \in H^0(\mathbb{P}_{\mathbb{F}}^N, \mathcal{O}(1))$ such that $\mathcal{C} \cap V(\pi, \bar{\ell}) = e_1 \bar{x}_1 + \dots + e_r \bar{x}_r$ where $\bar{x}_1, \dots, \bar{x}_r \in \mathbb{A}_{\mathbb{F}}^N \subset \mathbb{P}_{\mathbb{F}}^N$ are distinct points with $[\mathbf{k}(\bar{x}_i) : \mathbb{F}] = f_i$. Then there is an open neighborhood of linear forms $\ell \in H^0(\mathbb{P}_{\mathcal{O}}^N, \mathcal{O}_{\mathbb{P}^N}(1))$ such that $\ell \equiv \bar{\ell} \pmod{\pi}$ and $\mathcal{C}_K \cap V(\ell) = x_1 + \dots + x_r$ where $x_1, \dots, x_r \in \mathcal{C}_K$ are distinct points and $\mathbf{k}(x_i)/K$ has ramification degree e_i and residual degree f_i .*

The following criteria for when a quotient of a regular ring is also regular will be helpful in the proof.

Lemma 4.3. *Let $(R, \mathfrak{m})$ be a 2-dimensional regular local ring. For any $\ell \in \mathfrak{m} \setminus \mathfrak{m}^2$, the quotient $R/(\ell)$ is a 1-dimensional regular local ring.*

Proof. Since $\ell \in \mathfrak{m} \setminus \mathfrak{m}^2$, there is some element $z \in \mathfrak{m}$ such that $\mathfrak{m} = \langle \ell, z \rangle$. Thus, the lemma follows from [SP, Lemma 10.106.3]. $\square$

Proof of Proposition 4.2. Let B be the $\mathcal{O}$ -algebra such that $\mathcal{C} \cap \mathbb{A}_{\mathcal{O}}^N = \text{Spec } B$. We may choose homogeneous coordinates $z_0, \dots, z_N$ on $\mathbb{P}_{\mathcal{O}}^N$ so that $\bar{x}_1, \dots, \bar{x}_r \in D(z_0) \subset \mathbb{P}_{\mathbb{F}}^N$ and $\bar{\ell} = z_N$. This in particular allows us to treat $z_1, \dots, z_N$ as well-defined elements of B .

Let $\mathfrak{m}_1, \dots, \mathfrak{m}_r \subset B$ be the maximal ideals corresponding to $\bar{x}_1, \dots, \bar{x}_r$. We have $(\pi, z_N) \subset \mathfrak{m}_i$ for all i . Since $\mathcal{C} \rightarrow \text{Spec } \mathcal{O}$ is smooth and $\text{Spec } \mathcal{O}$ is regular, $B_{\mathfrak{m}_i}$ is a regular local ring of dimension 2 [SP, Lemma 10.163.10]. In addition, since $\mathcal{C}_{\mathbb{F}}$ is a smooth $\mathbb{F}$ -scheme, we have that $(B/(\pi))_{\bar{\mathfrak{m}}_i} \simeq B_{\mathfrak{m}_i}/\pi B_{\mathfrak{m}_i}$ is a regular local ring of dimension 1. Hence, by [SP, Lemma 10.106.4], there exists an element $g_i \in B_{\mathfrak{m}_i}$ such that $(\pi, g_i)B_{\mathfrak{m}_i} = \mathfrak{m}_i B_{\mathfrak{m}_i}$. Since $z_N \in \mathfrak{m}_i$, we can write $z_N = \alpha_i \pi + \beta_i g_i$ for some $\alpha_i, \beta_i \in B_{\mathfrak{m}_i}$.

We will show that there exists a linear form $\ell' = a_0 + a_1 z_1 + \dots + a_{N-1} z_{N-1}$ (equivalently a choice of $a_0, \dots, a_{N-1} \in \mathcal{O}$) such that for $\ell = z_N + \pi \ell'$, we have that $(B/\ell)_{\bar{\mathfrak{m}}_i}$ is a regular local ring of dimension 1 (i.e., a discrete valuation ring [SP, Lemma 10.119.7]) for all $1 \leq i \leq r$. By Lemma 4.3, this is equivalent to

$$\ell = z_N + \pi(a_0 + a_1 z_1 + \dots + a_{N-1} z_{N-1}) = \beta_i g_i + \pi(\alpha_i + a_0 + a_1 z_1 + \dots + a_{N-1} z_{N-1}) \notin \mathfrak{m}_i^2 B_{\mathfrak{m}_i}, \quad (4.2)$$

for all i . (Note that since ℓ is a combination of z_N and π , it is already in $\mathfrak{m}_i B_{\mathfrak{m}_i}$ for each i .) Since π, g_i form a basis for $(\mathfrak{m}_i B_{\mathfrak{m}_i})/(\mathfrak{m}_i^2 B_{\mathfrak{m}_i})$ as an $\mathbb{F}$ -vector space, $\ell \in \mathfrak{m}_i^2 B_{\mathfrak{m}_i}$ if and only if $\beta_i \in \mathfrak{m}_i B_{\mathfrak{m}_i}$ and

$$(\alpha_i \bmod \mathfrak{m}_i) + a_0 + a_1(z_1 \bmod \mathfrak{m}_i) + \dots + a_{N-1}(z_N \bmod \mathfrak{m}_i) \in \mathfrak{m}_i B_{\mathfrak{m}_i}. \quad (4.3)$$

Equation 4.3 only depends on the reduction of $a_0, \dots, a_{N-1}$ in $\mathcal{O}/(\pi) = \mathbb{F}$, so the images of solutions $a_0, \dots, a_{N-1}$ form a translate of a hyperplane in $\mathbb{F}^N$. Since $\#\{i \leq r : \beta_i \in \mathfrak{m}_i B_{\mathfrak{m}_i}\} \leq r < \#\mathbb{F}$, the union of r translates of hyperplanes cannot cover $\mathbb{F}^N$. Hence, there exists a choice of $a_0, \dots, a_{N-1} \in \mathcal{O}$ whose reduction modulo π satisfies (4.3) for all i for which $\beta_i \in \mathfrak{m}_i B_{\mathfrak{m}_i}$. Indeed, since the condition is on the reduction modulo π , there is an open neighborhood $U \subset \mathcal{O}^N$ consisting of $a_0, \dots, a_{N-1}$ that satisfy this condition. Setting $\ell = \ell_{\mathbf{a}} = z_N + \pi(a_0 + a_1 z_1 + \dots + a_{N-1} z_{N-1})$, we have shown that $B_{\mathfrak{m}_i}/(\ell B_{\mathfrak{m}_i})$ is a discrete valuation ring for all $\mathbf{a} \in U$.

To simplify notation, let $A := B/(\ell)$. Write $\bar{\mathfrak{m}}_i$ for the image of $\mathfrak{m}_i$ in A . By [SP, Proposition 10.9.14], $A_{\bar{\mathfrak{m}}_i} \simeq B_{\mathfrak{m}_i}/(\ell B_{\mathfrak{m}_i})$, and so $A_{\bar{\mathfrak{m}}_i}$ is a discrete valuation ring with maximal ideal $\bar{\mathfrak{m}}_i$. In particular, $A_{\bar{\mathfrak{m}}_i}$ has a unique nonzero prime ideal (i.e., $\bar{\mathfrak{m}}_i$), so we can refer to *the* residue field of $A_{\bar{\mathfrak{m}}_i}$, and every nonzero ideal is a power of $\bar{\mathfrak{m}}_i$.

Since $\ell \in \mathfrak{m}_i$ we have $\mathbf{k}(\bar{x}_i) = B/\mathfrak{m}_i \simeq A/\bar{\mathfrak{m}}_i \simeq A_{\bar{\mathfrak{m}}_i}/\bar{\mathfrak{m}}_i A_{\bar{\mathfrak{m}}_i}$. Thus the residue field of $A_{\bar{\mathfrak{m}}_i}$ is $\mathbf{k}(\bar{x}_i)$. To prove that $\pi A_{\bar{\mathfrak{m}}_i} = \bar{\mathfrak{m}}_i^{e_i} A_{\bar{\mathfrak{m}}_i}$, it suffices to show that e_i is the smallest power of the image of $\bar{\mathfrak{m}}_i$ in $A_{\bar{\mathfrak{m}}_i}/(\pi A_{\bar{\mathfrak{m}}_i})$ that is 0. Repeatedly using the fact that localization respects quotients [SP, Proposition 10.9.14] and also that $\ell \equiv z_N \pmod{\pi}$, we have

$$A_{\bar{\mathfrak{m}}_i}/(\pi A_{\bar{\mathfrak{m}}_i}) \simeq (A/\pi A)_{\bar{\mathfrak{m}}_i} \simeq B_{\mathfrak{m}_i}/(\pi, \ell) B_{\mathfrak{m}_i} = B_{\mathfrak{m}_i}/(\pi, z_N) B_{\mathfrak{m}_i}.$$

By assumption, $\text{Spec}(B_{\mathfrak{m}_i}/(\pi, z_N) B_{\mathfrak{m}_i})$ is nonreduced of order e_i , and so $\bar{\mathfrak{m}}_i^{e_i} = 0$ but $\bar{\mathfrak{m}}_i^{e_i-1} \neq 0$. All together, we have shown that the $\mathcal{O}$ -algebra $A_{\bar{\mathfrak{m}}_i}$ is a discrete valuation ring with fraction field $\mathbf{k}(\bar{x}_i)$ and for which the valuation of π is exactly e_i .

Finally, let $\mathfrak{p}_i \subset A$ be the (unique) height 0 prime ideal corresponding to $(0) \subset A_{\bar{\mathfrak{m}}_i}$. Then, since $\mathfrak{p}_i A_{\mathfrak{p}_i} = 0$, $A_{\mathfrak{p}_i}$ is the fraction field of $A_{\bar{\mathfrak{m}}_i}$. The generic fiber of the subscheme of $\mathcal{C} \cap V(\ell)$ corresponding to the prime $\mathfrak{p}_i$ is therefore a point $x_i \in C_K \cap V(\ell)$ with $\mathbf{k}(x_i) = A_{\mathfrak{p}_i} = \text{Frac}(A_{\bar{\mathfrak{m}}_i})$. We conclude that $\mathbf{k}(x_i)/K$ has ramification degree e_i and residual degree f_i . $\square$

4.2. Proof of Theorem 4.1. Under the assumptions, we have an embedding $\mathcal{C} \hookrightarrow \mathbb{P}_{\mathcal{O}}^n$ for $n = h^0(C, D) - 1$, and on the special fiber, the linear equivalence in (4.1) guarantees that there exists a linear form in $H^0(\mathbb{P}_{\mathbb{F}}^n, \mathcal{O}_{\mathbb{P}_{\mathbb{F}}}^n(1))$ vanishing exactly on the divisor $e_1 x_1 + \dots + e_r x_r$. Since $\mathbb{F}_v$ is sufficiently large, the points $x_1, \dots, x_r$ are contained in some affine n -space within $\mathbb{P}^n$. Thus, Proposition 4.2 yields an open set $U \subset H^0(\mathbb{P}_{\mathcal{O}}^N, \mathcal{O}_{\mathbb{P}^N}(1))$ of linear forms ℓ such that $V(\ell) \cap C \subset \mathbb{P}_{k_v}^N$ consists of r closed points whose residue fields have ramification degree e_i and residual degree f_i . Since k is dense in k_v , there exists a k -rational linear form $\ell \in U$. By projecting away from a suitable linear subspace of $V(\ell)$, we obtain a k -rational map $\pi: C \rightarrow \mathbb{P}^1$ with $C \cap V(\ell) = \pi^{-1}(\tilde{t})$ for some $\tilde{t} \in \mathbb{P}^1(k)$ [VV26, Lemma 3.1.2]. Since $\ell \in U$, $\mathbf{k}(\pi^{-1}\tilde{t})/k$ has splitting profile $\mathbf{P}$ at v . Then by an extension of Hilbert irreducibility (see, e.g., [BCLV, Lemma 5.2]), there exists a point $t \in \mathbb{P}^1(k)$ such that $x := \pi^{-1}(t)$ is integral (i.e., a point) and has the desired v -adic splitting behavior. $\square$

4.3. Examples of likely splitting profiles. In this section, we show that certain splitting profiles are likely under very mild conditions. For example, if the curve C has a rational point x , then total ramification is likely in $|ex|$ for sufficiently large e (Corollary 4.9(1)). More generally, if $\mathbf{P}^0$ is likely in a complete linear system $|D|$, then every sufficiently large multiple $\mathbf{P} = d\mathbf{P}^0 := \{(de^0, f^0) : (e^0, f^0) \in \mathbf{P}^0\}$ is likely in $|dD|$ (Section 4.3.1). This allows us to prove a strong converse to Theorem 1.1 (Corollary 4.9). At the other extreme, unramified splitting profiles of sufficiently large degree are also likely (Section 4.3.2). In Section 4.3.3, we show how these results combine to prove Proposition 1.10.

4.3.1. Sufficiently large multiples of likely splitting profiles are likely.

Lemma 4.4. *Let C be a nice curve over a number field k and let $C \rightarrow \mathbb{P}^1$ be a degree d morphism. For any finite collection of degree d extensions $k_1, \dots, k_n$ over k , there exists a point $t \in \mathbb{P}^1(k)$ such that C_t is integral and $\mathbf{k}(C_t) \not\subset k_i$ for any $1 \leq i \leq n$.*

Proof. For all sufficiently large v , there exists a k_v point on a nontrivial étale twist of the Galois closure $C \rightarrow \mathbb{P}^1$. In other words, for all sufficiently large v , there exists a point $t = t(v) \in \mathbb{P}^1(k)$ such that $\mathbf{k}(C_t)_v$ is a nontrivial étale k_v -algebra. If we additionally constrain v so that it splits completely in k_i for all i , then $\mathbf{k}(C_t) \not\subset k_i$ as desired. $\square$

Lemma 4.5. *Let C be a nice curve over a number field k , let $x \in C$ be a closed point, and let v be a nonarchimedean place of k . If $|x|$ is basepointfree, then there exist infinitely many closed points $y \in C$ such that $y \sim x$ and $\mathbf{P}_v(\mathbf{k}(y)/k) = \mathbf{P}_v(\mathbf{k}(x)/k)$.*

Proof. If $|x|$ is basepointfree then there exists an effective divisor $D \in |x|$ whose support is disjoint from x . Thus there is a function $f: C \rightarrow \mathbb{P}^1$ of degree d such that $f^*0 = x$ and $f^*\infty = D$. Now choose a $t_0 \in \mathbb{P}^1(k)$ such that t_0 is not a branch point of f and $\mathbf{k}(f^*t_0)$ is not isomorphic to $\mathbf{k}(x)$, which exists by Lemma 4.4. Pick a place $w \neq v$ such that $\mathbf{k}(f^*t_0) \otimes k_w \not\subset \mathbf{k}(x) \otimes k_w$ (there are infinitely many such possibilities, since the places that split completely in $\mathbf{k}(x)$ characterize the field extension). Then by an extension of Hilbert

irreducibility [BCLV, Lemma 5.2] there exists a closed point $t \in \mathbb{P}^1(k)$ such that the fiber $y := f^*t$ is integral and

$$\mathbf{k}(y) \otimes k_v \simeq \mathbf{k}(x) \otimes k_v \quad \mathbf{k}(y) \otimes k_w \not\simeq \mathbf{k}(x) \otimes k_w.$$

In particular, $y \sim x$, $y \neq x$, and $P_v(\mathbf{k}(x)/k) = P_v(\mathbf{k}(y)/k)$.

To obtain infinitely many such points, we iteratively add more conditions at more places to ensure that each new point y_i is different than those previously obtained. $\square$

Lemma 4.6. *Let C be a nice curve over a number field k , let $x \in C$, let v be a nonarchimedean place of k where C has good reduction and such that $\text{Spec } \mathcal{O}_{\mathbf{k}(x)} \rightarrow \mathcal{C}$ is injective over v , and let d be a positive integer. If $h^0(C, dx) = h^0(C_{\mathbb{F}_v}, (dx)_{\mathbb{F}_v})$ and $|(dx)_{\mathbb{F}_v}|$ is very ample, then there exists a closed point $y \in C$, linearly equivalent to dx such that*

$$P_v(\mathbf{k}(y)/k) = d P_v(\mathbf{k}(x)/k).$$

Proof. By Lemma 3.1, there are distinct points $\bar{x}_1, \dots, \bar{x}_r \in C_{\mathbb{F}_v}$ such that

$$\deg(\bar{x}_i) = f_i \quad \forall i, \quad \text{and} \quad \sum_i e_i \bar{x}_i \sim x_{\mathbb{F}_v}.$$

After multiplying both sides by d , we can apply Theorem 4.1 to obtain a closed point $y \in C$ such that $\mathbf{k}(y)$ has splitting profile $d P_v(\mathbf{k}(x)/k)$ at v . $\square$

Proposition 4.7. *Let C be a nice curve over a number field k , let D be a divisor of positive degree, and let P^0 be a splitting profile with $\deg(P^0) = \deg(D)$. If P^0 is likely in $|D|$, then for every sufficiently large d , the splitting profile $P = d P^0$ is likely in $|dD|$.*

Proof. Let $\mathcal{S} = \{v : \exists x \in |D| \text{ integral, } P_v(\mathbf{k}(x)/k) = P^0\}$; by assumption $\mathcal{S}$ is infinite and so there exists an $x \in |D|$ integral with $P_v(\mathbf{k}(x)/k) = P^0$ for some $v \in \mathcal{S}$.

First we consider the case when $|D| = \{x\}$, so $\mathcal{S} = \{v : P_v(\mathbf{k}(x)/k) = P^0\}$ implies that P^0 is unramified. Let $d > 2g(C)/\deg(D)$ so that dD is very ample. Since any smooth projective embedding extends to one over $\mathcal{O}_{k,S}$ for some finite set S of primes, $|dD|_{\mathbb{F}_w}$ is very ample for all $w \notin S$. Further, by upper semicontinuity, there is a finite set of primes S' such that $h^0(C, dD) = h^0(C_{\mathbb{F}_w}, (dD)_{\mathbb{F}_w})$ for all $w \notin S'$. Finally, since $\text{Spec } \mathcal{O}_{\mathbf{k}(x)} \rightarrow \mathcal{C}$ is generically injective it is injective above w for all places w away from a finite set S'' . Thus, applying Lemma 4.6, we conclude that

$$\mathcal{S} \setminus (S \cup S' \cup S'') \subset \{v : \exists y \in |dD|, P_v(\mathbf{k}(y)/k) = d P^0\}.$$

Since $\mathcal{S}$ is infinite and S, S', S'' are all finite, this gives the desired conclusion.

Now assume $|D| \not\subseteq \{x\}$. Since x is integral, this implies $|D| = |x|$ is basepoint free. Thus, by Lemma 4.5, for any d , and any place v , we may find distinct closed points $y_1, \dots, y_d \in |x|$ such that $P_v(\mathbf{k}(y_i)/k) = P_v(\mathbf{k}(x)/k)$ and $x \neq y_i$ for all i . Since dx and $y_1 + y_2 + \dots + y_d$ are contained in $|dx|$ and have disjoint support, we may find a morphism $f: C \rightarrow \mathbb{P}^1$ of degree equal to $d \cdot \deg(x)$ where the fiber above 0 is the union of all y_i . Then by [BCLV, Lemma 5.2] there exists a $t \in \mathbb{P}^1(k)$ such that the fiber $z := C_t \in |dx|$ is integral and $P_v(\mathbf{k}(z)/k) = \cup_i P_v(\mathbf{k}(y_i)/k) = d P^0$. Thus $d P^0$ is likely in $|dD|$. $\square$

Corollary 4.8. *Let C be a nice curve over a number field k and let $x \in C$ be a closed point. If v is unramified in $\mathbf{k}(x)$, then for every sufficiently large d , the splitting profile $d P_v(\mathbf{k}(x)/k)$ is likely in $|dx|$.*

Proof. By the Chebotarev density theorem, the splitting profile $P_v(\mathbf{k}(x)/k)$ is likely in $|x|$. The result therefore follows from Proposition 4.7. $\square$

Corollary 4.9. *Let C be a nice curve over a number field k .*

- (1) *If $x \in C(k)$, then total ramification is likely in $|dx|$ for all sufficiently large d .*
- (2) *If $\phi: C \rightarrow \mathbb{P}^1$ is totally ramified at a point $z \in C(\bar{k})$, then total ramification is likely in $|d\phi^*\infty|$ for all sufficiently large d .*

Proof. The first statement follows from Corollary 4.8. The second statement will follow from Proposition 4.7 once we prove that total ramification is likely in $|\phi^*\infty|$. The map ϕ spreads out to a morphism of good reduction away from finitely many places (see, e.g. [BCLV, Lemma 5.1]). For each v in the positive density set of places for which $z \in C(k_v)$ and for which ϕ has good reduction, we may find a $t \in \mathbb{P}^1(k)$ v -adically close to $\phi(z)$ and such that C_t is integral, so that $\mathbf{k}(C_t)$ will be totally ramified at v (see, e.g., [BCLV, Lemma 5.2 and Theorem 1.1]). Thus, total ramification is likely in $|\phi^*\infty|$. $\square$

4.3.2. *Sufficiently large unramified splitting profiles are likely.*

Proposition 4.10. *Let C be a nice curve over a number field k , let D be a very ample divisor. Then for every unramified splitting profile P with $\deg(P) = \deg(D)$ the set*

$$\{v \in \Omega_k : \exists y \in C, y \sim D, P_v(\mathbf{k}(y)/k) = P\}$$

contains all sufficiently large primes.

Proof. The case when $\deg(D) = \deg(P) = 1$ is either trivial (when $C = \mathbb{P}^1$) or vacuous, so henceforth we assume $d := \deg(D) \geq 2$. Since $|D|$ is assumed to be very ample, the incidence correspondence $I \subset C \times |D|$ is an S_d cover of $|D|$ [ACGH85, Lemma, Page 111]. By Hilbert irreducibility, we can find a line in $L \subset |D|$ such that $I_L \rightarrow L$ remains an S_d cover. Thus by the Chebotarev density theorem, Hensel's Lemma, and Hilbert irreducibility for every sufficiently large prime w , there exists a $t \in L(k)$, such that I_t is integral and has splitting profile P at w (see, e.g. [BCLV, Lemma 5.2 and Theorem 1.1]). Then the projection of I_t onto C is a point $y \in C$ with $\deg(y) = \deg(I_t)$, hence $y \simeq I_t$ as k -schemes. In particular, y also has splitting profile P at w . $\square$

4.3.3. *Proof of Proposition 1.10.* The statement is trivial when $\deg(P) = 1$. If $\deg(P) = 2$, then $\phi_{|2x|}: E \rightarrow \mathbb{P}^1$ is a double cover ramified at ∞ . Thus, for all but finitely many places v , there exists $t \in \mathbb{P}^1(k)$, v -adically close to ∞ such that E_t is integral and $\mathbf{k}(E_t)$ is totally ramified at v . Further, Hensel's Lemma implies that for all but finitely many places v , there exists $t_1, t_2 \in \mathbb{P}^1(k)$ such that E_{t_i} is integral, $\mathbf{k}(E_{t_1})$ is split at v , and $\mathbf{k}(E_{t_2})$ is inert at v . Thus Proposition 1.10 holds for all degree 2 splitting profiles. Henceforth, we assume that $\deg(P) > 2$ and so $\deg(P)x$ is very ample.

Assume that all ramification degrees in P are equal, i.e., that $\mathbf{e} = (d, d, \dots, d)$ for some fixed positive integer. Thus, we may think of P as a multiple of an unramified profile P' , i.e.,

$$\mathbf{f}(P) = \mathbf{f}(P'), \quad \mathbf{e}(P') = (1, \dots, 1), \quad \mathbf{e}(P) = d\mathbf{e}(P')$$

In this case, the desired statement follows from Proposition 4.10 and Corollary 4.8.

If there are at least two distinct ramification degrees in P , then we may write the profile as a disjoint union of profiles $P^{(1)}, \dots, P^{(n)}$ each with a fixed ramification degrees. By the argument in the previous paragraph, for every sufficiently large prime v , we have a point

$y_i \in E$ such that $\mathbf{k}(y_i)/k$ has splitting profile $\mathbf{P}^{(i)}$ at v . Since $\mathbf{P}^{(i)} \neq \mathbf{P}^{(j)}$ for $i \neq j$, $y_i \neq y_j$. Thus we may apply Lemma 4.11 below iteratively to obtain a point $y \in E$ with splitting profile $\mathbf{P} = \mathbf{P}^{(1)} \cup \dots \cup \mathbf{P}^{(n)}$ at v . $\square$

Lemma 4.11. *Let $D^{(1)}, D^{(2)}$ be divisors on C such that their sum is basepoint free and let $\mathbf{P}^{(1)}, \mathbf{P}^{(2)}$ be splitting profiles with $\deg(D^{(i)}) = \deg(\mathbf{P}^{(i)})$. Fix a nonarchimedean place v . If there exist points $x^{(i)} \in |D^{(i)}| \subset C$ with $\mathbf{P}_v(\mathbf{k}(x^{(i)})/k) = \mathbf{P}^{(i)}$ and $x^{(1)} \neq x^{(2)}$, then there exists a point $y \in |D^{(1)} + D^{(2)}|$ with splitting profile $\mathbf{P}^{(1)} \cup \mathbf{P}^{(2)}$ at v .*

Proof. Since $x^{(1)} \in |D^{(1)}|$ and $x^{(2)} \in |D^{(2)}|$, the sum $x^{(1)} + x^{(2)}$ is an effective divisor in $|D^{(1)} + D^{(2)}|$. Further, since $|D^{(1)} + D^{(2)}|$ is basepoint free, there exists a morphism $f: C \rightarrow \mathbb{P}^1$ with $\deg(f) = \deg(D^{(1)}) + \deg(D^{(2)})$ such that $f^*0 = x^{(1)} + x^{(2)}$. Since $x^{(1)} \neq x^{(2)}$, f is unramified at 0 so in particular, the fiber is an étale algebra. By weak approximation combined with Hilbert irreducibility (see, e.g., [BCLV, Lemma 5.2]), we may find a $t \in \mathbb{P}^1(k)$ such that $y := f^*t$ is integral and has $\mathbf{k}(y) \otimes_{k_v} \simeq \mathbf{k}(x^{(1)}) \otimes_{k_v} \times \mathbf{k}(x^{(2)}) \otimes_{k_v}$. In particular, y has splitting profile $\mathbf{P}^{(1)} \cup \mathbf{P}^{(2)}$ at v . $\square$

5. SETS OF PLACES REALIZING A GIVEN SPLITTING PROFILE

The goal of this section is to prove Corollary 5.6 (which implies Theorem 1.11), i.e., to understand the density set of places realizing a maximal likely splitting profile. To do so, we need to understand how the converse of Lemma 3.2 can fail.

Definition 5.1. Let $H \subset \text{Pic}_C^0$ abelian subvariety and D a divisor of positive degree. Fix a splitting profile $\mathbf{P}$ with $\deg(D) = \deg(\mathbf{P})$. Define $Y_{\mathbf{P}, D, H} \subset \text{Sym}_C^{\mathbf{f}(\mathbf{P})}$ to be the preimage of $H + [D]$ under the composition

$$\text{Sym}_C^{\mathbf{f}(\mathbf{P})} \xrightarrow{\text{Abel-Jacobi}} \text{Pic}^{\mathbf{f}(\mathbf{P})} \xrightarrow{[e(\mathbf{P})]} \text{Pic}_C^{\deg(\mathbf{P})}$$

Lemma 3.2 states that if a splitting profile $\mathbf{P}$ is likely in $|D + H|$, then $Y_{\mathbf{P}, D, H} \neq \emptyset$. However, the converse does fail. For example, if $x, y \in C(k)$ are such that $|x + y| = \{x + y\}$ and $\mathbf{P} = \{(1, 2)\}$, then $Y_{\mathbf{P}, [x] + [y], 0} \neq \emptyset$, but since $|x + y|$ consists of the single reducible divisor $x + y$, the splitting profile $\mathbf{P}$ is not realized by a single place. The issue is that the point of $Y_{\mathbf{P}, [x] + [y], 0}$ really arises from the profile $\mathbf{P}' := \{(1, 1), (1, 1)\}$. To explore this further, we fix some notation.

Remark 5.2. The proof of Theorem 1.11 will be in the case when $H = 0$.

Definition 5.3. The splitting profile $\mathbf{P}'$ **covers** the splitting profile $\mathbf{P}$, denoted $\mathbf{P}' \succ \mathbf{P}$, if for some $e, e', f, f' \in \mathbb{Z}_{>0}$ we have one of the following conditions

- (1) $\mathbf{P}' \setminus (\mathbf{P} \cap \mathbf{P}') = \{(e, f), (e, f')\}$, and $\mathbf{P} \setminus (\mathbf{P} \cap \mathbf{P}') = \{(e, f + f')\}$, or
- (2) $\mathbf{P}' \setminus (\mathbf{P} \cap \mathbf{P}') = \{(e + e', f)\}$, and $\mathbf{P} \setminus (\mathbf{P} \cap \mathbf{P}') = \{(e, f), (e', f)\}$.

The covering relation $\succ$ generates a partial order $\geq$ on the set of splitting profiles: we say that $\mathbf{P}' \geq \mathbf{P}$ if there is a sequence of covering relations, $\mathbf{P}' := \mathbf{P}^{(n)} \succ \dots \succ \mathbf{P}^{(1)} = \mathbf{P}$. If $\mathbf{P}' \geq \mathbf{P}$, then $\deg(\mathbf{P}') = \deg(\mathbf{P})$, $\sum_{e_i \in \mathbf{e}(\mathbf{P}')} e_i \geq \sum_{e_i \in \mathbf{e}(\mathbf{P})} e_i$ and $\sum_{f_i \in \mathbf{f}(\mathbf{P}')} f_i \leq \sum_{f_i \in \mathbf{f}(\mathbf{P})} f_i$. Additionally, any splitting profile $\mathbf{P}$ sits between a totally ramified splitting profile and a profile with maximum residual degree:

$$\{(\deg(\mathbf{P}), 1)\} \geq \mathbf{P} \geq \{(1, \deg(\mathbf{P}))\}.$$

Lemma 5.4. *Let $P' > P$ be splitting profiles. Then there is a commutative diagram*

$$\begin{array}{ccc} \mathrm{Sym}_C^{f(P')} & \xrightarrow{\text{Abel-Jacobi}} & \mathrm{Pic}_C^{f(P')} \\ \downarrow & & \searrow [e(P')] \\ \mathrm{Sym}_C^{f(P)} & \xrightarrow{\text{Abel-Jacobi}} & \mathrm{Pic}_C^{f(P)} \xrightarrow{[e(P)]} \mathrm{Pic}_C^{\deg(P)} \end{array}$$

If (1) holds, then the vertical maps are given by partial summation and the leftmost map is finite surjective. If (2) holds, then the vertical maps are given by a partial diagonal and are closed immersions. In particular, for every abelian subvariety $H \subset \mathrm{Pic}_C^0$ and every divisor D of positive degree with $\deg(D) \mid \deg(P)$, we have a finite morphism

$$Y_{P',D,H} \rightarrow Y_{P,D,H}.$$

Proof. It suffices to prove this in the case that $P \cap P' = \emptyset$. In that case, the first three statements of the lemma follow from the two diagrams ((1) on the left and (2) on the right):

$$\begin{array}{ccc} \mathrm{Sym}_C^{(f,f')} & \longrightarrow & \mathrm{Pic}_C^{(f,f')} \\ \downarrow + & & \downarrow + \\ \mathrm{Sym}_C^{f+f'} & \longrightarrow & \mathrm{Pic}_C^{f+f'} \xrightarrow{[e]} \mathrm{Pic}_C^{\deg(P)} \end{array} \quad \begin{array}{ccc} \mathrm{Sym}_C^f & \longrightarrow & \mathrm{Pic}_C^f \\ \downarrow \Delta & & \downarrow \Delta \\ \mathrm{Sym}_C^{(f,f)} & \longrightarrow & \mathrm{Pic}_C^{(f,f)} \xrightarrow{[(e,e')]} \mathrm{Pic}_C^{\deg(P)} \end{array}$$

which evidently have the desired properties. Since $Y_{P',D,H}$ and $Y_{P,D,H}$ are both complete preimages of $[D] + H$, the last claim follows from the previous ones. $\square$

With this framework, we can determine the extent to which a converse of Lemma 3.2 holds.

Proposition 5.5. *Let D be a very ample divisor and let P be a splitting profile with $\deg(D) = \deg(P)$. If $Y_{P,D,\{0\}} \neq \emptyset$ then there exists a splitting profile $P' \geq P$ such that P' is likely in $|D|$.*

The converse of this proposition holds by Lemmas 3.2 and 5.4. Before proving the proposition, we first record the following corollary, which easily implies Theorem 1.11.

Corollary 5.6. *Let D be a very ample divisor on a nice curve C/k and let P be a splitting profile with $\deg(D) = \deg(P)$. Assume that P is maximal among splitting profiles that are likely in $|D|$, i.e., assume that P is likely in $|D|$, and that for all $P' > P$, the profile P' is unlikely in $|D|$. Then the set*

$$\mathcal{T}_{P,D} := \{v \in \Omega_k : \exists x \in C, x \sim D \text{ with } P_v(\mathbf{k}(x)/k) = P\} \quad (5.1)$$

is infinite and almost equal to the set $\{v : Y_{P,D,\{0\}}(\mathbb{F}_v) \neq \emptyset\}$. Furthermore, $\mathcal{T}_{P,D}$ is Frobenian with positive density. It has density 1 if and only if $Y_{P,D,\{0\}}$ is pseudo-split.

Remark 5.7. In order to extend Corollary 5.6 to other splitting profiles, one needs to have a deep understanding of when the existence of an $\mathbb{F}_v$ point on $Y_{P,D,\{0\}}$ is the limit of a closed point $x \in C$ with $P > P_v(\mathbf{k}(x)/k)$. For example, let $C \subset \mathbb{P}^2$ be the smooth plane quartic defined by $(z_0^2 + z_1^2)^2 + z_0 z_2(z_1^2 + z_2^2) = 0$. The line $z_0 = 0$ meets C at the single point $[0 : 0 : 1]$ with multiplicity 4, which shows that $Y_{\{(4,1)\},K_C,\{0\}}(\mathbb{F}_v) \neq \emptyset$ for all v . Additionally, the line $z_2 = 0$ meets C at the degree 2 point $V(z_0^2 + z_1^2, z_2)$ with multiplicity 2. Thus, for any v that splits in $\mathbb{Q}(i)$, we have $Y_{\{(2,1),(2,1)\},K_C,\{0\}}(\mathbb{F}_v) \neq \emptyset$. Lemma 3.2 shows that these

$\mathbb{F}_v$ -points of $Y_{\{(2,1),(2,1)\},K_C,\{0\}}$ (the ones corresponding to the bitangent $z_2 = 0$) will never arise as the limit of a closed point $x \in |K_C|$ with splitting profile $\{(2,2)\}$. However, the $\mathbb{F}_v$ -points of $Y_{\{(4,1)\},K_C,\{0\}}$ can arise as such a limit. Indeed, that happens for $v = 5$ and $x = C \cap V(z_0 - 50z_2)$.

Proof of Theorem 1.11. Since $J_C(k)$ has rank 0, there are finitely many k -divisor classes $[D_1], \dots, [D_n]$ of degree d , which are guaranteed to be very ample by the Riemann–Roch theorem. If $\mathbf{P} := \{(d, 1)\}$ is likely in $|D_i|$, we apply Corollary 5.6 with $D = D_i$, which is valid since there are no splitting profiles $\mathbf{P}' > \mathbf{P}$. If $\mathbf{P}$ is unlikely in $|D_i|$, then $\mathcal{T}_{\mathbf{P},D_i}$ is finite, hence Frobenian. All together, this implies the result since the finite union of Frobenian sets is Frobenian. $\square$

Proof of Corollary 5.6. The second statement follows from the first by work of Loughran–Smeets and Loughran–Skorobogatov–Smeets, specifically [LS16, Lemma 3.9] and [LSS20, Paragraph before Proposition 2.11]. It remains to prove the first statement. Since we work with a fixed divisor D and $H = \{0\}$, for any splitting profile $\mathbf{P}'$ with $\deg(\mathbf{P}') = \deg(D)$, we set $Y_{\mathbf{P}'} := Y_{\mathbf{P}',D,\{0\}}$.

Fix $v \in \mathcal{T}_{\mathbf{P},D}$. With only finitely many exceptions, we may apply Lemma 3.2 and conclude that $Y_{\mathbf{P}}(\mathbb{F}_v) \neq \emptyset$. Now let v be a nonarchimedean place such that $Y_{\mathbf{P}}(\mathbb{F}_v) \neq \emptyset$, in other words we have effective divisors $z_1, \dots, z_r$ (where $r := r(\mathbf{P})$) such that $\deg(z_i) = f_i$ and $\sum_i e_i[z_i] = [D_{\mathbb{F}_v}]$. Let $\mathbf{P}'$ be a splitting profile with $\mathbf{P}' > \mathbf{P}$. Then, by assumption, $\mathbf{P}''$ is unlikely for all $\mathbf{P}'' \geq \mathbf{P}'$. Thus, by Proposition 5.5, $Y_{\mathbf{P}'} = \emptyset$. In particular,

$$\bigcup_{\mathbf{P}' > \mathbf{P}} \{v \in \Omega_k : Y_{\mathbf{P}'}(\mathbb{F}_v) \neq \emptyset\}$$

is finite. Hence, away from finitely many places, the collection $(z_i) \in Y_{\mathbf{P}}$ is not in the image of an $\mathbb{F}_v$ -point on $Y_{\mathbf{P}'}$ for any $\mathbf{P}' > \mathbf{P}$. In other words, each z_i is integral and (after possibly excluding finitely many more places) we may apply Theorem 4.1 to conclude that $v \in \mathcal{T}_{\mathbf{P},D}$. $\square$

Proof of Proposition 5.5. If $Y_{\mathbf{P},D,\{0\}} \neq \emptyset$, then $Y_{\mathbf{P},D,\{0\}}$ has closed points. Fix some closed point y and set $L := \mathbf{k}(y)$, which is a finite extension of L . Thus, over L , $Y_{\mathbf{P},D,\{0\}}$ has a rational point which, by definition, corresponds to effective divisors $Z_1, \dots, Z_{r(\mathbf{P})}$ (defined over L) where $\deg(Z_i) = f_i$ and $\sum_i e_i Z_i \simeq D$. We claim that after possibly replacing $\mathbf{P}$ with a splitting profile greater than it, we may assume that each Z_i is integral and $Z_i \neq Z_j$ for all $i \neq j$. Indeed, if $Z_i = \sum_j n_{i,j} Z'_{i,j}$, then the effective divisors $(Z'_{i,j})_{i,j}$ give an L -point on $Y_{\mathbf{P}',D,\{0\}}$ where

$$\mathbf{P}' := \bigcup_{i=1}^{r(\mathbf{P})} \bigcup_j \{(e_i n_{i,j}, \deg(Z'_{i,j}))\}.$$

By definition of the ordering, $\mathbf{P}' \geq \mathbf{P}$. Further, if some $Z'_{i,j} = Z'_{i',j'}$ for $(i, j) \neq (i', j')$, then we may replace $\mathbf{P}'$ with one that covers it using the relation in Definition 5.3(2).

Henceforth, we assume that each Z_i is integral and set $\mathbf{P}' := \cup_{(e,f) \in \mathbf{P}} \{(e, 1)\}_{j=1}^f$. Now let v be any place of k that is split completely in L and split completely in $\mathbf{k}(Z_i)/L$ for all i ; by the Chebotarev Density Theorem there are a positive density set of such places v . Further exclude the finitely many places where either C has bad reduction, where $Z_i \bmod v$ is nonreduced for some i , or where $Z_i \bmod v, Z_j \bmod v$ have overlapping support for some $i \neq j$. Then for any v in this positive density of places, the divisors Z_i give a k_v -point on $Y_{\mathbf{P}',D,\{0\}}$

that is not in the image of $Y_{P'', D, \{0\}}$ for any $P'' > P$. Furthermore, the reduction of these divisors Z_i give $\mathbb{F}_v$ -points satisfying the equation (4.1) from the hypothesis of Theorem 4.1. Since D is very ample, after possibly further excluding another finite set of places, all of the hypotheses of Theorem 4.1 will be satisfied and we deduce the existence of a point $x \in |D|$ such that $P_v(\mathbf{k}(x)/k) = P'$, as desired. $\square$

REFERENCES

- [ACGH85] E. Arbarello, M. Cornalba, P. A. Griffiths, and J. Harris, *Geometry of algebraic curves. Vol. I*, Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 267, Springer-Verlag, New York, 1985. MR0770932 [↑4.3.2](#)
- [BCLV] İrmak Balçık, Stephanie Chan, Yuan Liu, and Bianca Viray, *Number fields generated by points in linear systems on curves*, available at <https://arxiv.org/abs/2503.07846>. [↑1.1.3](#), [4.2](#), [4.3.1](#), [4.3.1](#), [4.3.1](#), [4.3.2](#), [4.3.3](#)
- [Bec91] Sybilla Beckmann, *On extensions of number fields obtained by specializing branched coverings*, J. Reine Angew. Math. **419** (1991), 27–53, DOI 10.1515/crll.1991.419.27. MR1116916 [↑1.1.3](#)
- [DG12] Pierre Dèbes and Nour Ghazi, *Galois covers and the Hilbert-Grunwald property*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 3, 989–1013, DOI 10.5802/aif.2714 (English, with English and French summaries). MR3013814 [↑1.1.3](#)
- [DL13] Pierre Dèbes and François Legrand, *Specialization results in Galois theory*, Trans. Amer. Math. Soc. **365** (2013), no. 10, 5259–5275, DOI 10.1090/S0002-9947-2013-05800-X. MR3074373 [↑1.1.3](#)
- [KLN19] Joachim König, François Legrand, and Danny Neftin, *On the local behavior of specializations of function field extensions*, Int. Math. Res. Not. IMRN **9** (2019), 2951–2980, DOI 10.1093/imrn/rny016. MR3947643 [↑1.1.3](#)
- [Leg17] François Legrand, *Hilbert specialization results with local conditions*, Rocky Mountain J. Math. **47** (2017), no. 6, 1917–1945, DOI 10.1216/RMJ-2017-47-6-1917. MR3725250 [↑1.1.3](#)
- [LMFDB] The LMFDB Collaboration, *The L-functions and modular forms database*, 2026. <https://www.lmfdb.org> [Online; accessed 2 June 2026]. [↑3.2.3](#)
- [LSS20] Daniel Loughran, Alexei N. Skorobogatov, and Arne Smeets, *Pseudo-split fibers and arithmetic surjectivity*, Ann. Sci. Éc. Norm. Supér. (4) **53** (2020), no. 4, 1037–1070, DOI 10.24033/asens.2439 (English, with English and French summaries). MR4157111 [↑5](#)
- [LS16] D. Loughran and A. Smeets, *Fibrations with few rational points*, Geom. Funct. Anal. **26** (2016), no. 5, 1449–1482, DOI 10.1007/s00039-016-0381-8. MR3568035 [↑5](#)
- [MR18] Barry Mazur and Karl Rubin, *Diophantine stability*, Amer. J. Math. **140** (2018), no. 3, 571–616, DOI 10.1353/ajm.2018.0014. With an appendix by Michael Larsen. MR3805014 [↑1.1.1](#)
- [PARI/GP] The PARI Group, *PARI/GP version 2.17.2*, Univ. Bordeaux, 2023. available from <http://pari.math.u-bordeaux.fr/>. [↑3.2.3](#)
- [Poo01] Bjorn Poonen, *Computing torsion points on curves*, Experiment. Math. **10** (2001), no. 3, 449–465. MR1917430 [↑3.2.3](#), [3.2.3](#)
- [Raw24] James Rawson, *Some obstructions to solvable points on higher genus curves*, J. Théor. Nombres Bordeaux **36** (2024), no. 3, 1009–1019, DOI 10.5802/jtnb.1304 (English, with English and French summaries). MR4890317 [↑1.12](#)
- [Ray83a] M. Raynaud, *Courbes sur une variété abélienne et points de torsion*, Invent. Math. **71** (1983), no. 1, 207–233, DOI 10.1007/BF01393342 (French). MR0688265 [↑2](#), [2.2](#)
- [Ray83b] ———, *Sous-variétés d’une variété abélienne et points de torsion*, Arithmetic and geometry, Vol. I, Progr. Math., vol. 35, Birkhäuser Boston, Boston, MA, 1983, pp. 327–352 (French). MR0717600 [↑2](#), [2.2](#)
- [SP] The Stacks project authors, *The Stacks project*, 2025. [↑4.1](#), [4.1](#)
- [VV26] Bianca Viray and Isabel Vogt, *Isolated and parameterized points on curves*, Essent. Number Theory **5** (2026), no. 1, 1–47, DOI 10.2140/ent.2026.5.1. MR5050108 [↑1.9](#), [\(i\)](#), [2.3](#), [2.4](#), [3.2.2](#), [4.2](#)

UNIVERSITY OF WASHINGTON, DEPARTMENT OF MATHEMATICS, BOX 354350, SEATTLE, WA 98195, USA

Email address: `bviray@uw.edu`

URL: `http://math.washington.edu/~bviray`

BROWN UNIVERSITY, DEPARTMENT OF MATHEMATICS, BOX 1917, 151 THAYER STREET, PROVIDENCE, RI 02912, USA

Email address: `ivogt.math@gmail.com`

URL: `https://www.math.brown.edu/ivogt/`